

MICRO ENTREPRENEUR LEDGER APPLICATION

AI-Powered Multi-Lingual Financial Management Platform for Small Businesses

PHASE - 1 REPORT

SUBMITTED BY

MANOJ KUMAR S 230701178

MOHAMED IKRAM KHD 230701188

KAMALESH L 230701136

MITHUN AR 230701186

*in partial fulfilment for the award of the degree
of*

**BACHELOR OF ENGINEERING IN
COMPUTER SCIENCE AND ENGINEERING**



**RAJALAKSHMI
ENGINEERING
COLLEGE**



**RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
ANNA UNIVERSITY, CHENNAI 600 025**

October 2026

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled "MICRO ENTREPRENEUR LEDGER APP: AI-POWERED MULTI-LINGUAL FINANCIAL MANAGEMENT PLATFORM FOR SMALL BUSINESSES" is the bonafide work of MANOJ KUMAR S (230701178), MOHAMED IKRAM KHD (230701188), KAMALESH L (230701136) and MITHUN AR (230701186), who carried out the work under my supervision. Certified further that, to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on these or any other candidates.

SIGNATURE

Dr. J. Manoranjini

Professor & Head

Department of Computer Science and Engineering,
Rajalakshmi Engineering College,
Thandalam, Chennai - 602105.

SIGNATURE

Ms. S. Sathiyavathi

Professor

Department of Computer Science and Engineering,
Rajalakshmi Engineering College,
Thandalam, Chennai - 602105.

Submitted to Project Viva-Voce Examination held on _____.

Internal Examiner

External Examiner

ABSTRACT

Micro-entrepreneurs such as Kirana shop owners, street vendors and small retailers handle a large number of low-value financial events every day: cash sales, UPI receipts, supplier purchases, customer credit, partial repayments and routine expenses. In many small businesses these records are still maintained in notebooks, informal message histories or scattered payment apps. The method is familiar, but it becomes difficult to search old entries, identify outstanding dues, reconcile supplier payments, or prepare a reliable summary of business activity.

The Micro Entrepreneur Ledger App is a mobile-first financial management platform designed to make these tasks easier without forcing the user to learn formal accounting terminology. The application combines a digital ledger, customer credit and partial repayment tracking, supplier accounts, invoice and payment-intimation generation, business reporting and multilingual voice-assisted transaction entry. English, Tamil and Tanglish speech can be converted into a structured transaction preview containing the transaction type, amount, category, associated party and date. The merchant remains in control because the parsed result is displayed for confirmation or editing before it is saved.

The Phase-1 implementation uses Next.js 14 with React 18 for the web interface, Next.js Server Actions and Route Handlers for server-side logic, Prisma ORM with SQLite for relational persistence, NextAuth.js with bcrypt-based password hashing for authentication, and the browser Web Speech API together with a custom heuristic NLP parser for speech-driven bookkeeping. Recharts supports financial visualisation, while CSV/PDF export, WhatsApp payment-intimation links and UPI-compatible payment data extend the ledger beyond basic transaction storage.

The implementation evidence in this report demonstrates the working login flow, multilingual voice assistant, transaction preview, digital ledger, supplier management, invoice/payment-intimation screens, report exports, business profile and financial insight dashboard. No unsupported benchmark accuracy is claimed for the voice engine in this phase; the displayed confidence score is treated as a per-command application indicator rather than a measured overall accuracy. The project aligns with Sustainable Development Goal 8 by supporting the digital productivity and financial organisation of small enterprises.

ACKNOWLEDGEMENT

Initially, we thank the Almighty for being with us through every stage of this work and for giving us the strength to complete the project and its documentation.

We express our sincere thanks to our Chairman Mr. S. Meganathan, B.E., F.I.E., our Vice Chairman Mr. Abhay Shankar Meganathan, B.E., M.S., and our respected Chairperson Dr. (Mrs.) Thangam Meganathan, Ph.D., for providing the infrastructure and academic environment required for our project work.

We also thank Dr. S. N. Murugesan, M.E., Ph.D., Principal, Rajalakshmi Engineering College, for the support and facilities provided to complete this work. We extend our sincere gratitude to Dr. J. Manoranjini, Professor and Head of the Department of Computer Science and Engineering, for her guidance and encouragement throughout the project.

We convey our sincere gratitude to our Project Guide, Ms. S. Sathiyavathi, Professor, Department of Computer Science and Engineering, for her continuous guidance, technical feedback and support during the planning, development and review of the Micro Entrepreneur Ledger App.

Finally, we thank the faculty members of the Department of Computer Science and Engineering, our friends and our families for their encouragement and support throughout the successful completion of this phase of the project.

TABLE OF CONTENTS

Section No.	Title	Page No.
CHAPTER 1	INTRODUCTION	1
1.1	General	1
1.2	Objective	2
1.3	Existing System	2
1.4	Proposed System	3
CHAPTER 2	LITERATURE SURVEY	5
2.1	Literature Review	5
2.2	Comparison and Discussion	6
2.3	Conclusion	6
CHAPTER 3	SYSTEM DESIGN	8
3.1	System Architecture	8
3.2	System Requirements	9
3.2.1	Software Requirements	9
3.2.2	Hardware Requirements	10
3.2.3	Data Requirements	10
3.2.4	Deployment and Scaling	10
3.3	System Design	11
3.3.1	Activity Diagram	11
3.3.2	Data Flow Diagram	13
3.3.3	Voice NLP Flow	13
3.3.4	Database Design	15
CHAPTER 4	PROJECT DESCRIPTION	16
4.1	Methodologies	16
4.2	Modules	16
4.2.1	Authentication & Business Profile Module	17
4.2.2	Digital Ledger & Transaction Module	17
4.2.3	Customer Credit / Khata Module	17
4.2.4	Supplier Management Module	18
4.2.5	Invoice & Payment Intimation Module	18
4.2.6	Voice Assistant & NLP Module	18
4.2.7	Reporting & Financial Insights Module	19
4.2.8	User Interface & Localization Module	19
CHAPTER 5	IMPLEMENTATION AND RESULT DISCUSSION	20
5.1	Implementation Results	20
5.1.1	Authentication and Business Setup	20
5.1.2	Ledger and Supplier Operations	21
5.1.3	Invoicing and Payment Intimation	22
5.1.4	Voice Command Processing	23

5.1.5	Reporting and Financial Insights	24
5.2	Functional Verification	25
5.3	Discussion	26
CHAPTER 6	CONCLUSION AND FUTURE WORK	27
6.1	Conclusion	27
6.2	Future Work	27
	REFERENCES	29

LIST OF TABLES

Table No.	Title	Page No.
Table 1	Comparative Analysis of Reviewed Literature	6
Table 2	Software Requirements	9
Table 3	Hardware Requirements	10
Table 4	System Module Summary	16
Table 5	Functional Verification Matrix	25

LIST OF FIGURES

Figure No.	Title	Page No.
Fig. 1	Overview of Proposed System	4
Fig. 2	Three-Tier Architecture Diagram	9
Fig. 3	Transaction Activity Diagram	12
Fig. 4	Data Flow Diagram	13
Fig. 5	Voice NLP Processing Flow	14
Fig. 6	Database Entity Relationship Summary	15
Fig. 7	Login and Demo Access Interface	20
Fig. 8	Business Profile and Sample Data Setup	21
Fig. 9	Digital Ledger Interface	21
Fig. 10	Supplier Management Interface	22
Fig. 11	Invoices and Payment Intimations	22
Fig. 12	Generated Payment Intimation	23
Fig. 13	Voice Assistant Listening Interface	23
Fig. 14	Voice Transaction Preview	24
Fig. 15	Reports and Tax Interface	24
Fig. 16	Financial Insights Dashboard	25

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
DFD	Data Flow Diagram
ER	Entity-Relationship
FK	Foreign Key
GST	Goods and Services Tax
HTML	HyperText Markup Language
NLP	Natural Language Processing
ORM	Object-Relational Mapping
PDF	Portable Document Format
PWA	Progressive Web Application
QR	Quick Response
RSC	React Server Components
SDG	Sustainable Development Goal
SQL	Structured Query Language
UI	User Interface
UPI	Unified Payments Interface
URI	Uniform Resource Identifier
UX	User Experience

CHAPTER 1

INTRODUCTION

1.1 General

Micro-enterprises form an important part of everyday commerce. Small shopkeepers, Kirana stores, street vendors, repair units, home-based businesses, tea stalls, fruit sellers and local retailers often operate with very limited administrative support. Their financial activity is frequent but fragmented across cash sales, UPI receipts, customer dues, supplier purchases, operating expenses and partial settlements. For many of these businesses, the most practical bookkeeping method is still a handwritten notebook or a collection of informal payment messages.

Paper ledgers are inexpensive and familiar, but they create operational problems as transaction volume grows. A business owner may need to locate an old credit entry, confirm whether a partial payment was recorded, compare today's expenses with sales, identify which supplier amount remains due, or prepare a simple profit view. In a notebook, these tasks require manual searching and repeated arithmetic. Records may be damaged, pages can be missed, and the owner has no automatic way to filter transactions or generate a consolidated view.

Research on small-business accounting and digital adoption shows that usability, perceived value and clear in-app guidance strongly influence whether small firms accept bookkeeping technology [3]-[5]. Studies of low-literacy and novice mobile users also show that complex text-heavy interfaces and unfamiliar terminology can become barriers, while graphical cues, local-language support and speech-based interaction can improve accessibility [1], [2]. These findings are directly relevant to a financial tool intended for busy merchants who may not want the complexity of full accounting software.

The Micro Entrepreneur Ledger App is designed as a simpler digital operating layer for daily business records. Rather than asking the merchant to think in formal debit-credit terminology, the application organizes familiar activities such as income, expense, purchase, customer credit, payment and supplier settlement. Searchable history, party-wise balances, invoices, reports and visual summaries allow the user to understand where money was received, where it was spent and what remains outstanding.

The current Phase-1 implementation is a full-stack web application built around Next.js 14 and React 18. Server Actions and Route Handlers execute business logic on the server, while Prisma ORM maps application objects to relational SQLite records. Authentication separates one merchant's records from another. A multilingual voice assistant uses the Web Speech API together with a custom heuristic parser to transform spoken English, Tamil or Tenglish phrases into a structured transaction preview. The merchant can confirm, edit or cancel the preview before any database change is made.

1.2 Objective

The primary objective of the project is to replace scattered or paper-based daily bookkeeping with a consistent digital ledger that remains understandable to a small-business owner. Each financial event should be stored with a date, transaction type, amount, category, payment mode and related customer or supplier whenever applicable. The ledger must remain searchable and filterable so the merchant can review previous activity without manually scanning pages.

A second objective is to make customer credit tracking reflect the way local retail actually works. Small customers often repay dues in irregular amounts rather than fixed installments. The system therefore treats every credit and every repayment as an individual ledger event and continuously derives the remaining balance. This allows a merchant to accept ₹100 today, ₹250 later and the balance after that without forcing an artificial EMI schedule.

The third objective is to reduce the effort needed to enter frequent transactions. The voice assistant accepts natural speech, uses browser speech recognition to obtain a transcript, and applies deterministic language, amount, intent, entity and date rules to build a structured transaction. The user is shown a preview before saving, which keeps human confirmation in the workflow and reduces the risk of an incorrect automatic write.

The fourth objective is to connect bookkeeping with day-to-day collection and communication. The invoicing module generates professional bills or payment intimations, supports UPI-compatible payment information and provides WhatsApp-ready communication. Supplier records, report export and financial summaries complete the basic business cycle instead of limiting the application to a simple list of transactions.

Finally, the project aims to remain lightweight and understandable. SQLite and Prisma provide a relational foundation without requiring a separate database server during Phase 1, while the unified Next.js codebase keeps the interface and server logic in one project. The system is aligned with SDG 8, particularly the goal of supporting entrepreneurship, productive activity and the growth of micro and small enterprises [11].

1.3 Existing System

The existing process for many target users is a mixture of paper notebooks, memory, messaging apps and separate payment histories. Even when a merchant accepts UPI, the payment application records the transfer but does not automatically classify it as business income, customer repayment or supplier settlement. The bookkeeping context remains outside the payment record.

Paper-based entries can be misplaced, damaged or written inconsistently, and searching older transactions is slow.

Outstanding customer credit is often calculated manually, especially when customers make partial repayments at irregular intervals.

Supplier purchases and payments may be written in a different notebook or remembered informally, making payable balances difficult to reconcile.

Payment applications provide transaction receipts but do not provide a complete business ledger, category view, customer balance or supplier history.

Conventional accounting systems are powerful but can introduce terminology and workflows that are unnecessarily complex for a very small shop.

Generating a payment reminder, invoice or financial summary usually requires a separate process rather than flowing directly from the ledger.

1.4 Proposed System

The proposed Micro Entrepreneur Ledger App brings the main financial activities of a small retail business into one interface. After authentication, the merchant can record and review income, expenses, purchases, customer credit, customer repayments and supplier payments. Customer and supplier records remain linked to transaction history so the application can calculate running balances and display the context behind each entry.

Voice entry is provided as an alternative to manual typing. Speech is transcribed by the browser Web Speech API and passed to a custom heuristic NLP pipeline. The parser identifies likely transaction intent, amount, category, person or entity and date. Instead of writing directly to the database, the system displays a transaction preview and asks the merchant to confirm or edit the values. The screenshot evidence supplied for this report shows a spoken command being converted to an Income entry of ₹150 with Product Sale as the category and a detected associated entity.

The project also includes supplier management, invoice and payment-intimation generation, report export, a business profile, language switching and a financial-insights dashboard. The current Phase-1 design is intentionally transparent: core ledger calculations and voice interpretation are rule-driven and auditable, while more advanced time-series forecasting remains future work. This avoids presenting unvalidated predictive accuracy as a finished capability.

Figure 1 presents the overall proposed system and shows how the user interface, business modules, server-side logic, database and communication/export functions are connected.

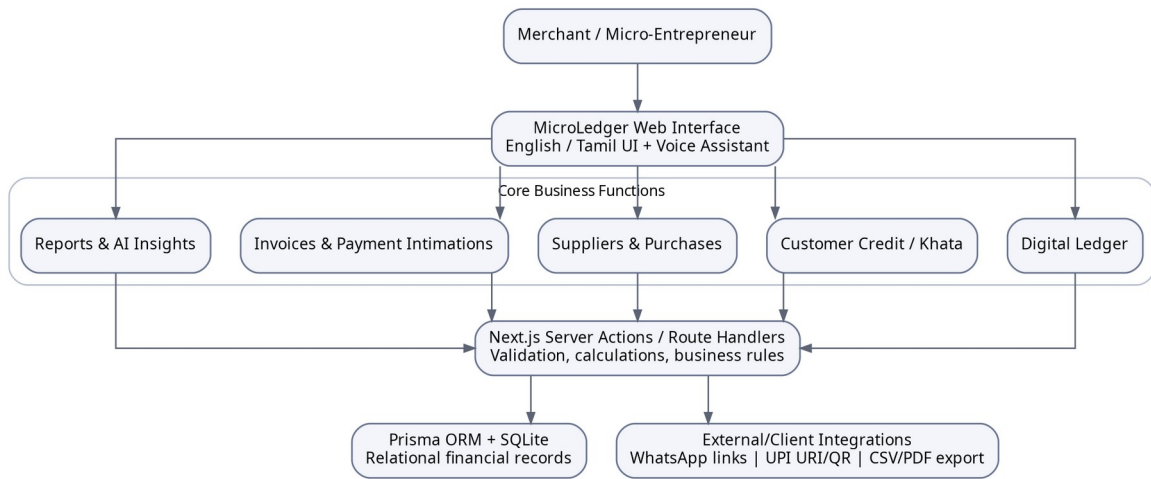


Fig. 1 - Overview of Proposed System

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Review

The literature relevant to this project falls into four closely related areas: usability for novice and low-literacy users, adoption of digital bookkeeping by small firms, digital payment adoption among merchants, and the technical foundations needed for a secure lightweight ledger. Together these areas explain why a small-business finance application must be simple to operate, tightly connected to real merchant workflows and cautious about introducing unnecessary accounting complexity.

Medhi et al. [1] studied mobile interfaces for novice and low-literacy users across developing-country contexts and found that standard text-based interfaces created significant usability problems. Their work emphasizes graphical cues, voice interaction and local-language support. Lalji and Good [2] similarly argue that technology for low-literacy populations should be developed around the users' context and through iterative, user-centred design rather than by copying interfaces made for highly literate users. These findings support the MicroLedger decision to use large visible actions, simple business terms and multilingual speech as an additional input method.

Small-business accounting research points to a different but complementary problem: the usefulness of an accounting system depends not only on correctness but also on whether the owner can understand and apply the information. Ritchi et al. [3] examined in-app navigation and guidance controls in a small-business accounting information system and linked these controls with users' domain understanding. Kumari [4] studied bookkeeping-application adoption among Indian SMEs and shopkeepers and reported that interactivity affects perceived value and adoption intention. Ylä-Kujala et al. [5] examined management-accounting adoption in small businesses and highlighted the challenges created by systems, resources and accounting know-how. These works justify an interface that exposes financial information without demanding specialist accounting knowledge.

Digital payments are already central to Indian retail, but payment acceptance alone does not solve bookkeeping. Ligon et al. [6] studied small merchants in Jaipur and found that adoption barriers are not explained only by access to infrastructure. More recent work by Motheram and Buteau [7] studies district-level UPI adoption and the relationship between merchant networks and digital infrastructure, while Rao and Upadhyaya [8] specifically examine UPI acceptance among small-scale merchants. These studies reinforce the importance of designing tools around the merchant's day-to-day workflow rather than assuming that the presence of UPI automatically creates organized financial records.

The IFC merchant-payments handbook [9] treats retail payments as part of a wider digital-finance ecosystem for micro and small enterprises. The Ministry of MSME Annual Report [10] documents the continuing importance of micro and small enterprises in India, while the United Nations SDG 8 framework

explicitly links entrepreneurship, technological upgrading and the growth of micro, small and medium enterprises with productive economic development [11]. The present project applies that broader policy direction at the level of a practical bookkeeping tool.

At the technical layer, the project uses established web and database components rather than experimental infrastructure. Next.js provides the full-stack App Router and server functions [13], Prisma provides a typed ORM layer over SQLite [14], and SQLite supports relational foreign-key constraints [15]. Authentication uses NextAuth/Auth.js concepts together with bcrypt-based password hashing, whose adaptive-cost design originates in the work of Provos and Mazières [16], [17]. The Web Speech API exposes speech recognition and synthesis capabilities in the browser [12]. These components make it possible to build the required workflow without a separate microservice architecture in Phase 1.

2.2 Comparison and Discussion

Table 1 - Comparative Analysis of Reviewed Literature

Literature Theme	Approach	Strength	Limitation	Relevance to This Project
Low-literacy mobile UI [1], [2]	Voice, graphical cues, user-centred mobile design	Improves accessibility and reduces dependence on complex text	Does not address bookkeeping logic or financial relationships	Use simple navigation, local-language support and voice-assisted entry
Small-business accounting systems [3], [5]	Accounting information systems, guidance controls, management accounting	Shows value of understandable financial information and in-app guidance	Often assumes structured accounting knowledge or formal business processes	Use familiar retail concepts: income, expense, purchase, credit and payment
Bookkeeping app adoption [4]	Interactive mobile bookkeeping application adoption	Directly relevant to SMEs and shopkeepers; highlights interactivity and perceived value	Focuses on adoption intention rather than end-to-end technical architecture	Combine low-friction input with immediately useful ledger, dues and reports
Merchant digital payments [6]-[8]	Merchant surveys, UPI adoption studies, digital-payment ecosystems	Explains merchant payment behaviour and importance of UPI	Payment acceptance alone does not organize business records	Link payments, customer/supplier context and bookkeeping in one ledger
Merchant payment ecosystem [9]	Digital financial-service and merchant-network practices	Connects payments with broader SME financial services	Provider-focused rather than merchant-led bookkeeping	Use UPI/WhatsApp as extensions of the ledger rather than separate products

The comparison shows a gap between payment technology and everyday financial organization. Research explains why merchants adopt digital payments, why novice users need simpler interfaces, and why small businesses benefit from usable accounting information. However, these areas are often treated separately. A merchant may have a UPI application without a usable ledger, or an accounting package without natural language entry and flexible Khata repayment. The Micro Entrepreneur Ledger App is positioned at this intersection: it combines simple record-keeping, party-wise credit, supplier tracking, voice-assisted input, invoice/payment intimation and reports in one merchant-oriented workflow.

2.3 Conclusion

The literature establishes a clear design direction for the project. The target users benefit from interfaces that reduce text complexity and support familiar interaction patterns [1], [2]. Small businesses require

financial systems that are understandable and useful in daily decision-making [3]-[5]. Indian merchant studies show that digital payment infrastructure is important but does not by itself solve adoption, workflow or record-management problems [6]-[9]. Based on these findings, the proposed system focuses on low-friction bookkeeping, relational financial records, flexible credit tracking, multilingual voice entry and direct payment communication rather than reproducing the full complexity of enterprise accounting software.

CHAPTER 3

SYSTEM DESIGN

3.1 System Architecture

The application follows a three-tier modern web architecture built around the Next.js 14 App Router. The Presentation Tier contains React Server Components and selected Client Components for forms, modals, charts and speech interaction. Tailwind CSS provides layout and styling, Framer Motion is used for interface transitions, Lucide supplies icons and Recharts renders financial visualisations. The voice assistant is a client-side interaction because microphone access and the Web Speech API are browser capabilities.

The Application and Business Logic Tier uses Next.js Server Actions and Route Handlers running on Node.js. These functions validate user input, verify the authenticated session, perform financial calculations, execute create/update/delete operations and request cache revalidation after mutations. Keeping these operations on the server prevents database logic from being exposed directly to the browser and avoids maintaining a separate Express backend for the current scale of the project [13].

The Data Persistence Tier uses Prisma ORM 7.9.x with SQLite. Prisma maps TypeScript models and relations to the relational database, while SQLite keeps the Phase-1 deployment self-contained. The project schema uses foreign-key relationships and referential actions to link merchants, customers, suppliers, transactions, invoices and related records. Prisma's SQLite connector and SQLite's foreign-key support provide an appropriate foundation for a single-node prototype [14], [15].

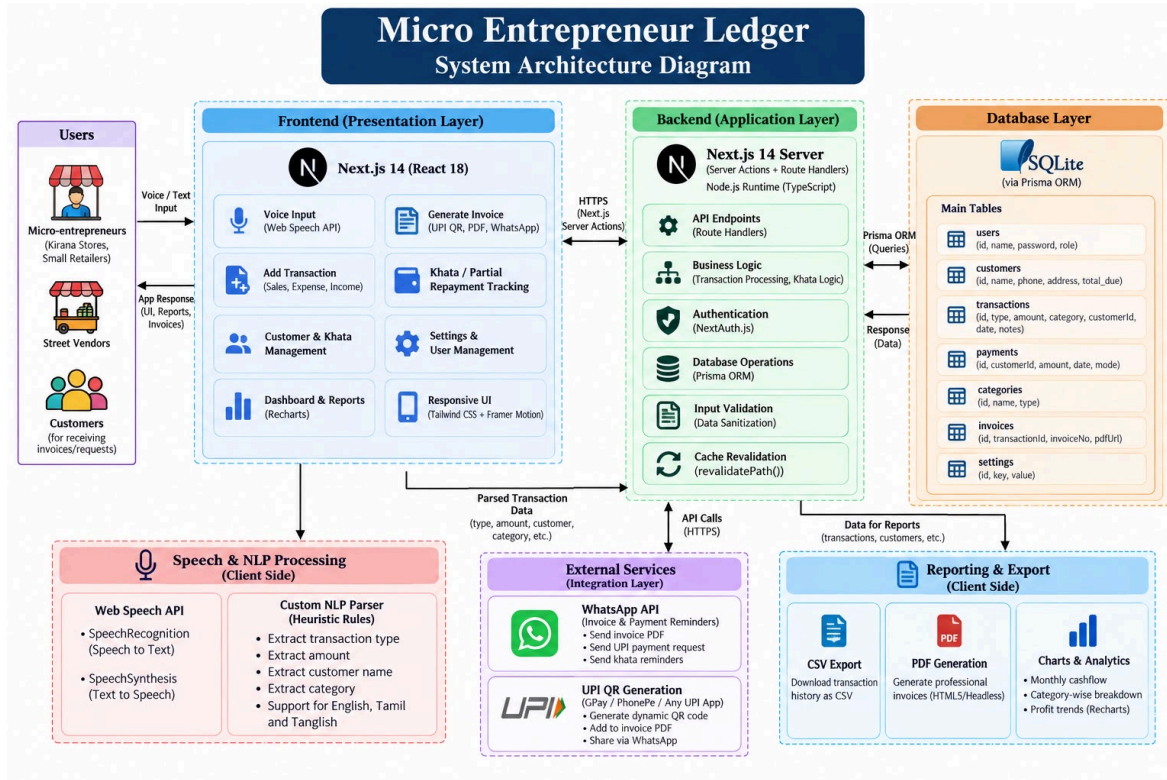


Fig. 2 - Three-Tier Architecture Diagram

3.2 System Requirements

The project is designed to run on ordinary student-development hardware and standard desktop/mobile browsers. No GPU or machine-learning training environment is required in Phase 1 because the current voice parser is deterministic and the financial insights are derived from recorded business data rather than from a trained forecasting model.

3.2.1 Software Requirements

Table 2 - Software Requirements

Software Component	Technology / Tool	Purpose
Operating System	Windows 10/11 or Linux	Development and local execution
Runtime / Framework	Node.js with Next.js 14 App Router	Full-stack server and web application
Frontend	React 18, Tailwind CSS, Framer Motion, Lucide React	Interface, responsive layout, animation and icons
Database / ORM	SQLite + Prisma ORM 7.9.x	Relational persistence and typed database access
Authentication	NextAuth.js / Auth.js patterns + bcryptjs	Session-based login and password hashing

Software Component	Technology / Tool	Purpose
Voice	Web Speech API + custom heuristic NLP	Speech recognition/synthesis and transaction parsing
Charts	Recharts	Financial trend and summary visualisation
Invoice / QR	qrcode.react + UPI-compatible URI data	QR-based payment information for invoices
Export	CSV generator, HTML5 print, PDF generation	Reports and printable records
Development Tool	Visual Studio Code	Source-code development and debugging

3.2.2 Hardware Requirements

Table 3 - Hardware Requirements

Hardware Category	Specification	Purpose in System
Processor	Dual-core 64-bit CPU or better	Runs the Next.js development/production server and local database
Memory	Minimum 4 GB RAM; 8 GB recommended	Supports Node.js, browser, editor and database during development
Storage	At least 2 GB free for project and dependencies	Source code, npm packages, SQLite database and exports
Microphone	Built-in or external microphone	Required only for voice-assisted entry
Display / Device	Desktop, laptop or modern mobile browser	Access to the responsive web interface
Network	Local network or Internet as required by deployment/browser speech service	Application access and browser-dependent speech recognition

3.2.3 Data Requirements

The system does not require an external training dataset in Phase 1. Its primary data is operational business data entered by the merchant: customer and supplier details, transaction type, amount, category, date, payment mode, invoices, repayment records, business profile information and audit information for voice commands. These records are relational rather than unstructured model-training data.

For demonstration and review, the application includes a sample Kirana dataset loader. The supplied implementation screenshot shows an option to populate the ledger with sample customers, suppliers and retail transactions so that reports and charts can be demonstrated without exposing real merchant data. This seeded sample data is used only as development/demo content and is not presented as evidence of real-world performance.

3.2.4 Deployment and Scaling

The current deployment is suitable for a single development machine, local shop system or small pilot. SQLite keeps the database in a local file and avoids the operational overhead of a separate database service. This is useful for Phase 1, but a public multi-merchant deployment with higher concurrency should move to a managed relational database such as PostgreSQL while retaining Prisma as the application data layer.

The early project proposal described an offline-first direction. The implemented Phase-1 system is better described as network-resilient when the Next.js server and SQLite database are hosted locally; it does not yet include a full client-side PWA synchronization layer. Browser speech recognition availability also depends on browser/device support. Full service-worker caching, IndexedDB transaction queues and conflict-aware background synchronization are therefore treated as Phase-2 work rather than claimed as completed functionality.

3.3 System Design

The detailed system design converts the high-level three-tier architecture into operational flows. The transaction activity diagram shows how manual and voice entry converge on the same preview-and-save path. The DFD shows how merchant actions reach the authentication, ledger, credit, supplier, invoice and report processes. A dedicated voice pipeline describes the deterministic parsing sequence, while the ER summary shows the principal database relationships documented in the project guide.

3.3.1 Activity Diagram

The main financial workflow begins with authentication and dashboard access. A merchant can either fill a transaction form manually or open the voice assistant. Voice input is transcribed and parsed before both paths converge on a transaction preview. Only after confirmation does the server validate and persist the record. This design deliberately places a human approval step between interpretation and financial mutation.

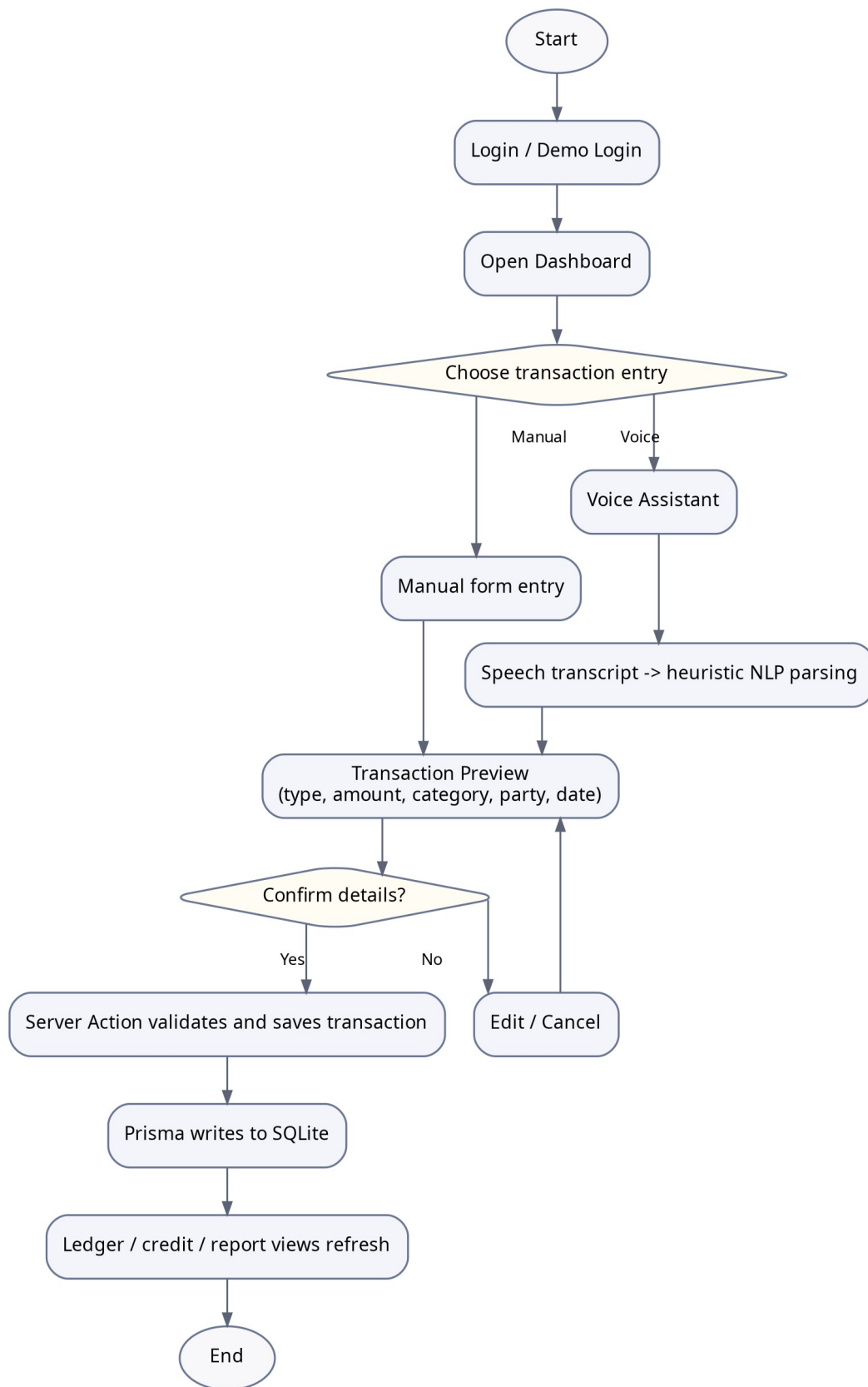


Fig. 3 - Transaction Activity Diagram

3.3.2 Data Flow Diagram

At Level 1, the system separates six major processes: authentication, ledger transactions, customer credit, supplier management, invoicing/payment intimation, and reporting/insights. Each process reads or writes the SQLite data store through Prisma. The invoicing process also produces customer-facing communication such as WhatsApp-ready payment intimations and UPI payment information.

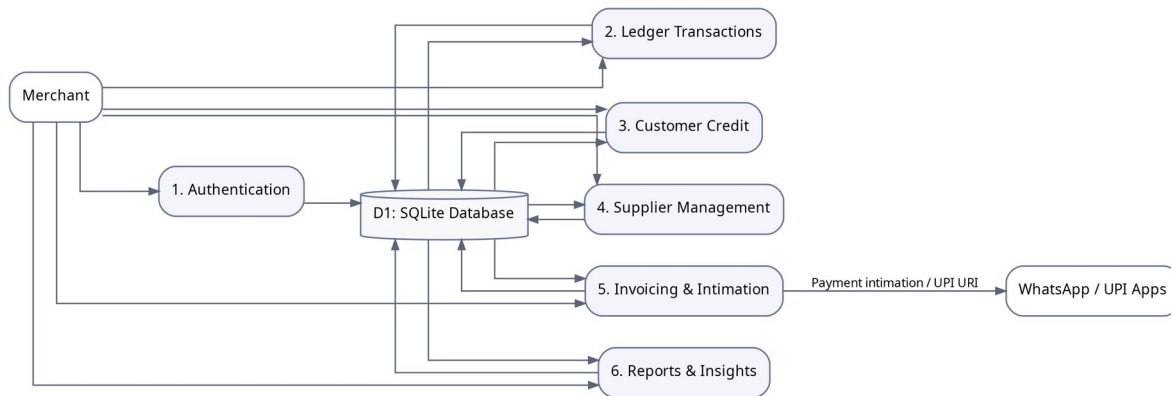


Fig. 4 - Data Flow Diagram

3.3.3 Voice NLP Flow

The voice engine is intentionally deterministic rather than dependent on a paid large-language-model API. After the Web Speech API returns a transcript, the parser checks the script/language context, extracts numeric amounts, classifies the financial intent, attempts to detect a customer or entity, normalizes the date and builds a structured object. Vocabulary rules cover English terms and common Tamil/Tanglish forms used in retail speech.

The output is not treated as final simply because the parser produced a result. The application displays the structured interpretation in a preview window together with an application confidence indicator. The user can confirm, edit or cancel. This confirmation stage is an important safety feature because speech recognition and colloquial language interpretation can be ambiguous.

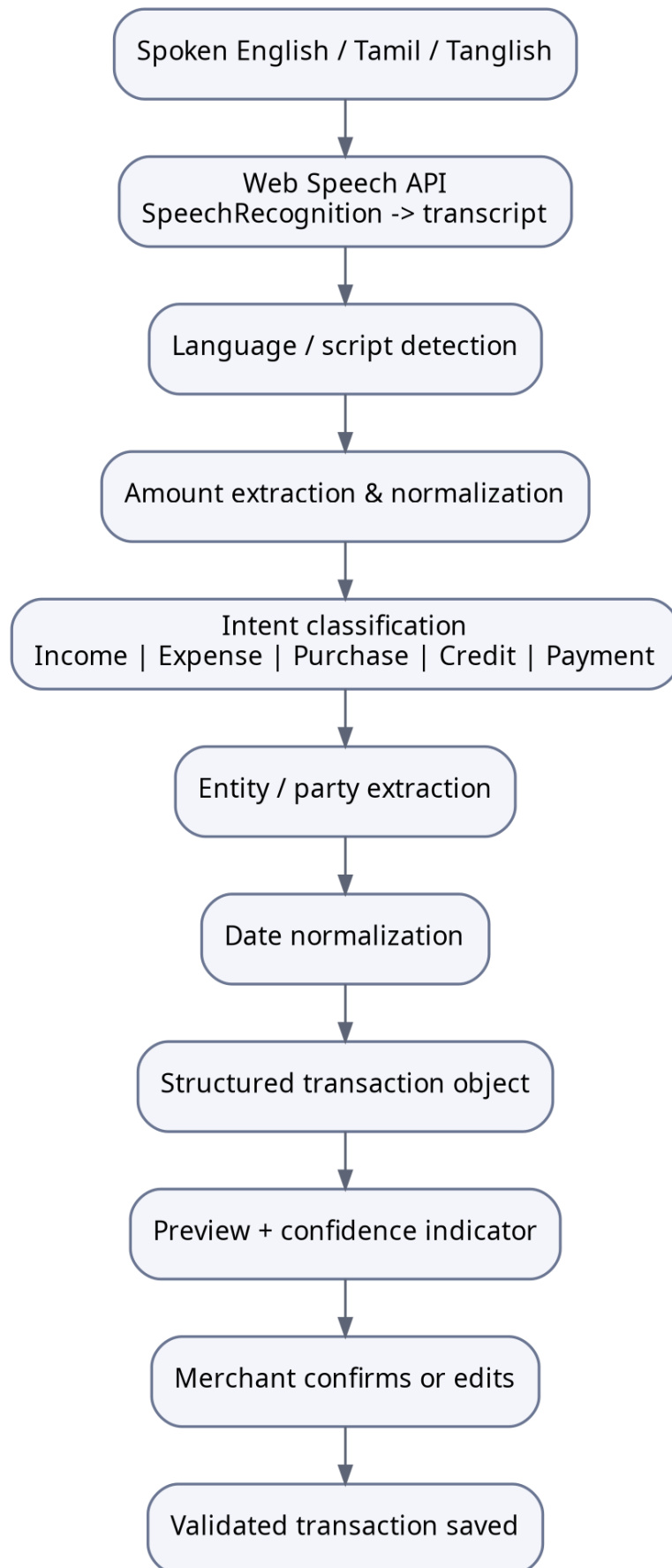


Fig. 5 - Voice NLP Processing Flow

3.3.4 Database Design

The Prisma schema described in the project guide contains eleven models: User, BusinessProfile, Customer, Supplier, Transaction, Invoice, InvoiceItem, EmiPlan, EmiInstallment, Goal and VoiceCommandAudit. User acts as the merchant account and owns the core business records. Transactions represent atomic financial movements such as income, expense, purchase, customer credit, customer payment and supplier payment. Optional customer and supplier relationships retain party context without forcing every transaction to belong to a party.

Invoice and InvoiceItem use a master-detail relation, while customer credit and repayment records can be associated with a customer and, where used, an EmiPlan/EmiInstallment structure. The project guide describes referential actions such as cascade deletion for child records and set-null behaviour where financial audit history should remain even after a contact is deleted. Prisma's relational model and SQLite foreign-key constraints provide the mechanisms needed for this design [14], [15].

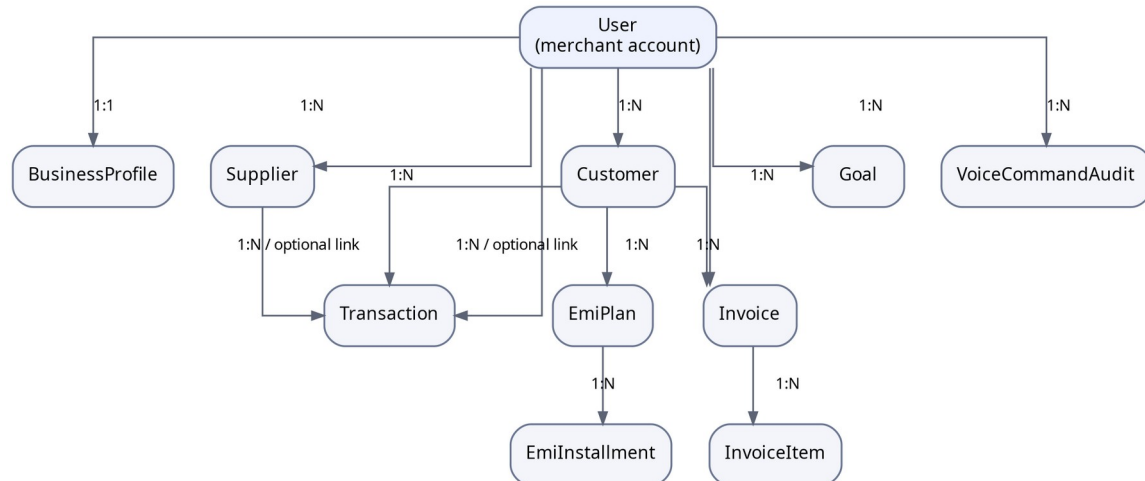


Fig. 6 - Database Entity Relationship Summary

CHAPTER 4

PROJECT DESCRIPTION

4.1 Methodologies

The project was developed using an iterative, feature-oriented software development approach. Requirements were first organized around the daily workflow of a small merchant: record money in and out, track customer dues, track supplier purchases, collect payments and understand the current position of the business. This prevented the implementation from being driven by technology alone.

Relational modelling was completed early because customer credit, supplier activity and invoices all depend on durable relationships between records. Authentication and protected routes were then established before business actions were added. Core transaction CRUD operations became the foundation for customer repayments, invoice settlement and reporting. This sequence reduced duplication because later modules reuse the same underlying financial records.

Voice processing was implemented as a separate interpretation layer in front of the existing transaction workflow. Instead of giving the voice parser its own database-writing path, the parser produces the same structured fields used by manual entry. The preview-confirmation step then routes the approved data into the standard transaction action. This keeps voice entry as an alternative input method rather than a second accounting system.

The user interface was refined around dark-mode panels, large primary actions, simple navigation and bilingual support. Development screenshots show a consistent left navigation with modules for Dashboard, Digital Ledger, Customer Credit, Invoices & Intimations, Suppliers, Income, Expenses, Reports & Tax, AI Insights and Business Profile. The same Voice Assistant action remains available throughout the authenticated interface.

4.2 Modules

Table 4 - System Module Summary

No.	Module	Main Responsibility
1	Authentication & Business Profile	Login, demo access, merchant profile, language/currency settings
2	Digital Ledger & Transactions	Unified income, expense, purchase, credit and payment history with filters/exports
3	Customer Credit / Khata	Party-wise credit and flexible partial repayment tracking
4	Supplier Management	Supplier purchases, payments, payable view and recent activity
5	Invoice & Payment Intimation	Invoice creation, due tracking, printable intimation and payment communication
6	Voice Assistant & NLP	English/Tamil/Tanglish speech input,

No.	Module	Main Responsibility
		parsing, preview and confirmation
7	Reporting & Financial Insights	Period reports, P&L/tax summary, CSV/PDF export and derived financial insights
8	UI & Localization	Responsive navigation, Tamil/English switching and merchant-oriented terminology

4.2.1 Authentication & Business Profile Module

Authentication provides the entry point to the private ledger. The implementation includes email/password login, a one-click Kirana demo login for review and a Google sign-in option in the interface. Password credentials are not stored as plaintext; the project uses bcryptjs hashing and session-based authentication through NextAuth.js. Bcrypt is designed as an adaptable-cost password hashing scheme, allowing the work factor to be increased as hardware becomes faster [16].

After login, the Business Profile module stores merchant-facing settings such as business name, business type, location, phone number, language and currency. The provided screenshot shows the Ramesh Kirana & General Stores demonstration profile and a one-click sample-data loader. Keeping business identity separate from financial transaction records simplifies invoice branding and future multi-merchant deployment.

4.2.2 Digital Ledger & Transaction Module

The Digital Ledger is the central financial record. It combines income, expenses, purchases, customer credit, customer payments and supplier payments in one searchable table rather than splitting the same cash movement across unrelated modules. Filters allow the merchant to focus on specific transaction types or time ranges, while CSV and PDF actions support export.

Each record contains a date, transaction type, category, party or note, payment mode and amount. Positive inflows and outgoing amounts are visually distinguished. Edit and delete actions are available where permitted. Other modules such as customer credit, supplier management and invoice settlement ultimately create or read ledger transactions, keeping financial summaries consistent.

4.2.3 Customer Credit / Khata Module

The customer credit module models informal retail credit as a running balance rather than a fixed banking-style repayment schedule. Total debt is derived from CUSTOMER_CREDIT transactions and repayments from CUSTOMER_PAYMENT transactions. The remaining amount is calculated as the difference between the two values. This directly supports real-world behaviour where a customer may repay any convenient amount at any time.

The project guide describes quick repayment actions and a clearance percentage used for visual progress. Because each payment remains an individual transaction, the business owner can inspect the sequence of credit and repayments instead of seeing only a final total. This approach also makes payment receipts and reminders easier to explain to the customer.

4.2.4 Supplier Management Module

Supplier management applies the same ledger principle to stock purchases and supplier settlements. A supplier card shows total purchases, amount paid, current payable status and recent activity. Recording a purchase creates an outgoing purchase transaction associated with the supplier; recording a payment updates the payment history and the amount still payable.

This is important for small retailers because supplier obligations may be spread across several wholesalers and dates. The implementation screenshot shows three sample suppliers with purchase and payment histories, demonstrating that payable information is derived from recorded activity rather than maintained as an isolated manual field.

4.2.5 Invoice & Payment Intimation Module

The invoicing module converts ledger and customer-balance information into a customer-facing document. The system tracks invoice number, billing date, due date, customer information, line description, quantity, rate, tax and total payable. An invoice can remain due or be marked as paid, allowing the financial state to be reflected back in the ledger.

For collection, the project supports UPI-compatible payment information and WhatsApp-ready intimations. NPCI documents QR-based UPI integration in which merchant information such as the UPI ID and amount can be carried in the QR payment data [20]. The application generates the payment information rather than operating a proprietary gateway; settlement is confirmed by the merchant and then recorded in the ledger.

4.2.6 Voice Assistant & NLP Module

The Voice Assistant is one of the project's main interaction features. It accepts spoken English, Tamil or Tanglish and converts the recognized transcript into a structured transaction proposal. The Web Speech API supplies recognition and speech-synthesis interfaces in supporting browsers [12]. Because browser implementations differ, the project treats speech as an optional input path rather than the only way to use the ledger.

The custom parser uses explicit rules rather than a remote LLM. It identifies language/script cues, normalizes numeric amounts, checks keyword sets for financial intent, extracts party names where possible and converts date phrases into application dates. This keeps the interpretation logic visible and inexpensive. A transaction preview then presents the result to the merchant before saving.

4.2.7 Reporting & Financial Insights Module

The reporting module groups ledger information by period and calculates revenue, expenses, stock purchases, estimated net profit and a tax/P&L summary. The interface offers preset periods such as Today, This Week, This Month, Last Month and All Time, along with CSV and PDF export. Recharts is used where charts are needed because it provides reusable React chart components [18].

The application also contains an AI Insights dashboard with components labelled Cash Flow Predictor, Profitability Assistant, Expense Leakage Detector and Savings Opportunity Dashboard. In this report these are described as financial insight features derived from available business data; no trained forecasting model or independently measured predictive accuracy is claimed for Phase 1. More advanced time-series prediction is reserved for future work.

4.2.8 User Interface & Localization Module

The interface uses a consistent dark visual system, persistent navigation and large high-contrast actions. English/Tamil switching is exposed in the header and voice dialog, while familiar business labels are used instead of formal accounting jargon wherever possible. This design direction is consistent with literature showing that novice and low-literacy users benefit from local-language support, graphical cues and reduced interaction complexity [1], [2].

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 Implementation Results

The Phase-1 system was implemented as a working full-stack web application rather than only as a conceptual prototype. The supplied screenshots show the authenticated merchant interface, working module navigation, sample financial records, supplier and invoice data, report exports and the voice transaction flow. These screenshots are used as implementation evidence in this chapter.

The results presented here are functional observations, not a statistical performance benchmark. No controlled user study or formal speech-recognition accuracy experiment was supplied for this report. Consequently, the report does not invent success percentages, latency claims or user-satisfaction values. Where the application displays a confidence score for a particular voice command, that value is reported only as a per-command interface output.

5.1.1 Authentication and Business Setup

The login interface provides a one-click Kirana merchant demonstration account as well as email/password login and a Google sign-in button. This allows a project reviewer to access a realistic merchant workspace without manually creating data before the demonstration.

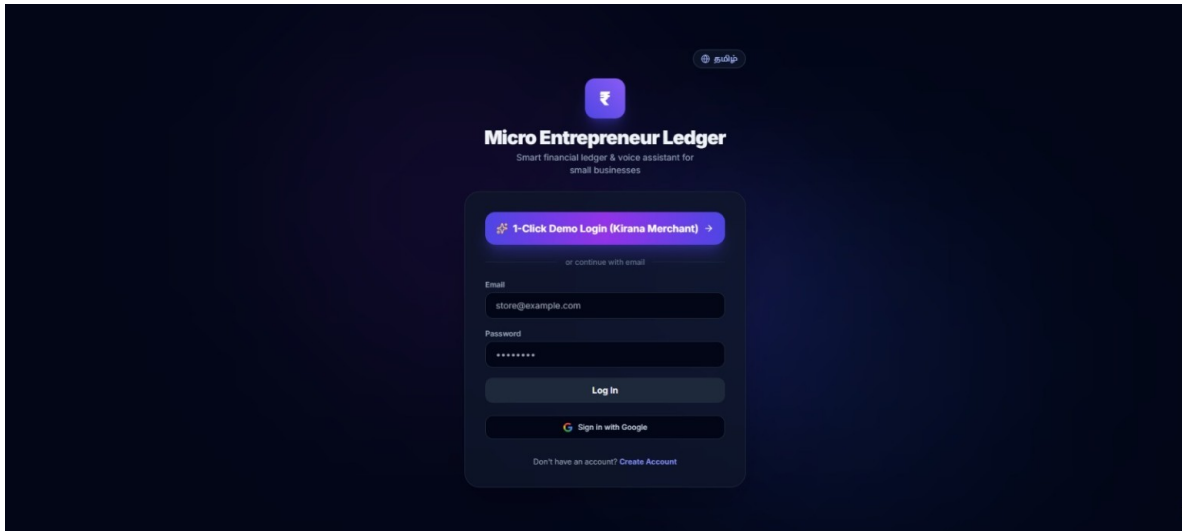


Fig. 7 - Login and Demo Access Interface

After authentication, the Business Profile page stores merchant identity and operating preferences. The implementation screenshot shows business name, Kirana/grocery-store type, Chennai location, phone number, English voice/UI language and Indian Rupee currency. A sample-data loader can pre-populate realistic demonstration records.

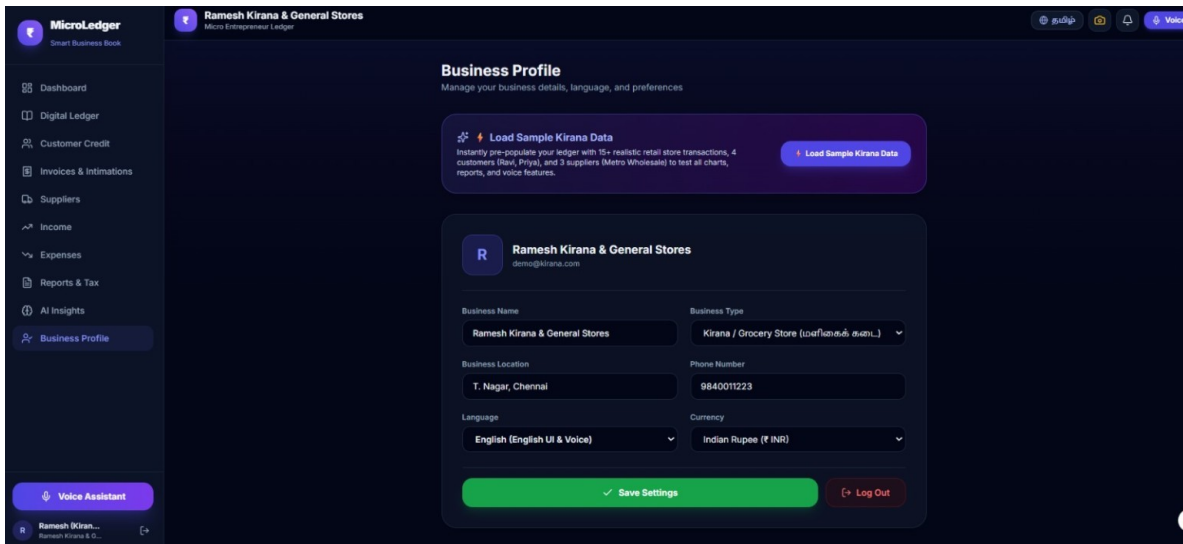


Fig. 8 - Business Profile and Sample Data Setup

5.1.2 Ledger and Supplier Operations

The Digital Ledger screenshot demonstrates a populated transaction table containing income, purchases and customer payments with date, category, party, payment mode and amount. Search, type filters, time-range controls and CSV/PDF export are visible in the same screen. This supports the project's goal of maintaining one searchable financial history rather than separate records for each business activity.

DATE	TYPE	CATEGORY	PARTY / NOTE	MODE	AMOUNT	ACTIONS
1 Oct 2028	INCOME	Product Sale	fried fried rice	CASH	+₹0	[Edit] [Delete]
9 Sept 2028	PURCHASE	Inventory	Metro Wholesale FMCG Purchase from Metro Wholesale FMCG	CASH	-₹1,000	[Edit] [Delete]
9 Sept 2028	CUSTOMER PAYMENT	Product Sale	Anitha Textiles Partial repayment received (Remaining balance: ₹1,400)	CASH	+₹200	[Edit] [Delete]
9 Sept 2028	PURCHASE	Inventory	Balaji Grains & Rice Mill Purchase from Balaji Grains & Rice Mill	CASH	-₹200	[Edit] [Delete]
9 Sept 2028	INCOME	Product Sale	—	UPI	+₹900	[Edit] [Delete]
9 Sept 2028	INCOME	Product Sale	—	CASH	+₹400	[Edit] [Delete]
9 Sept 2028	CUSTOMER PAYMENT	Product Sale	Priya Sundaram Partial repayment received (Remaining balance: ₹1,400)	CASH	+₹300	[Edit] [Delete]
9 Sept 2028	INCOME	Product Sale	—	CASH	+₹150	[Edit] [Delete]

Fig. 9 - Digital Ledger Interface

Supplier management is also implemented as a transaction-backed view. The supplier cards show total purchases, amount paid, payable state and recent activity, together with actions to record a payment or add a purchase. The sample data shows multiple suppliers, demonstrating that the module can maintain independent supplier histories within the same merchant account.

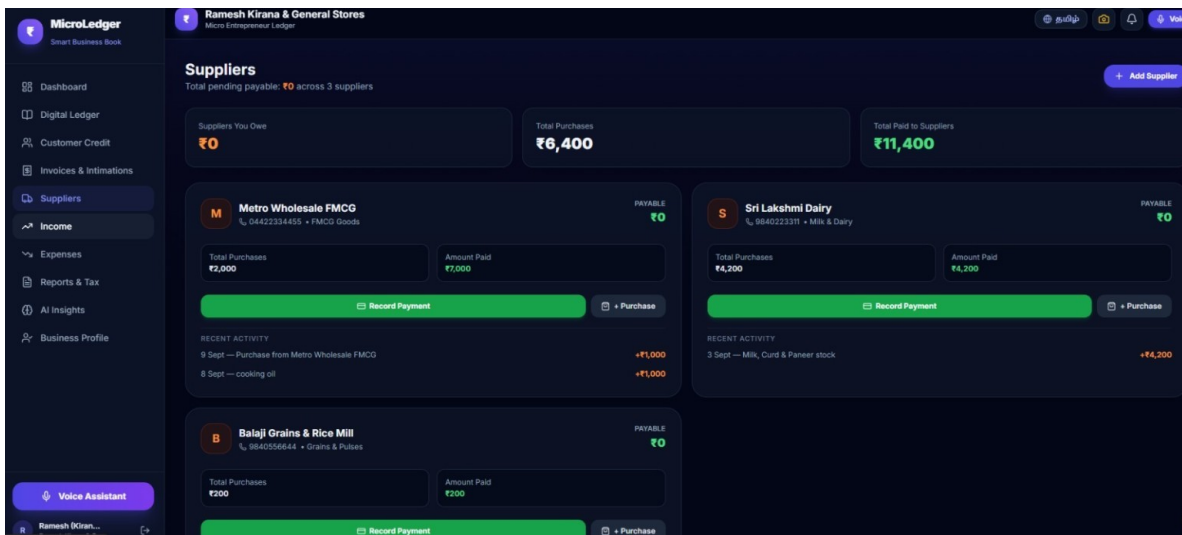


Fig. 10 - Supplier Management Interface

5.1.3 Invoicing and Payment Intimation

The invoice listing screen shows three issued payment intimations, their customer names, invoice numbers, dates, due dates, total amounts and payment status. Search and due/paid filters are available, along with actions to mark an invoice as paid or view it. This demonstrates the connection between customer dues and formal collection documents.

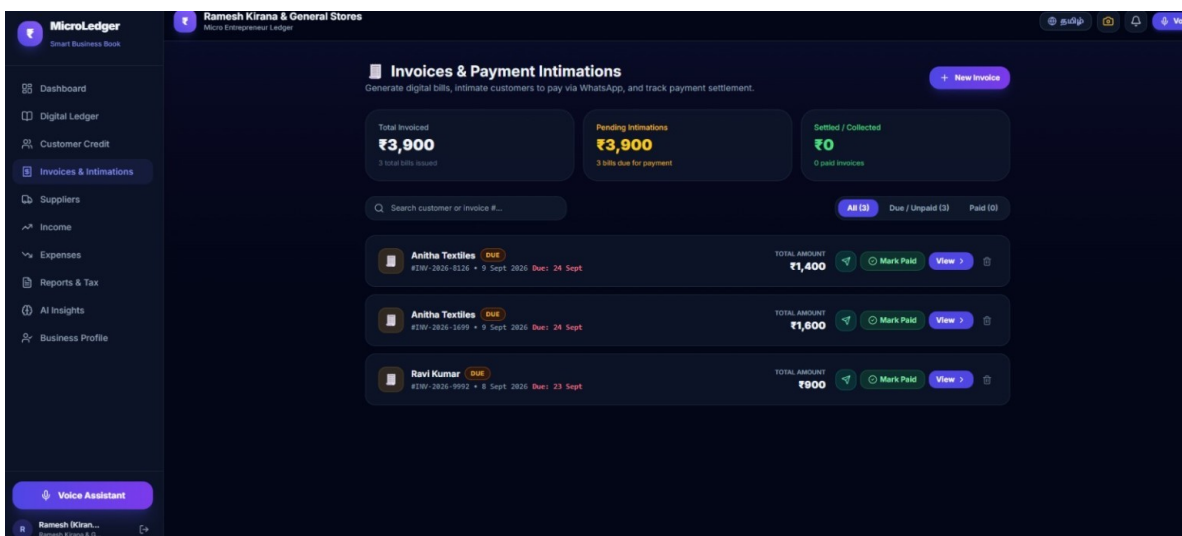


Fig. 11 - Invoices and Payment Intimations

The generated payment-intimation document provides customer details, the outstanding credit description, quantity, rate, tax percentage, amount, total payable, payment instructions, accepted payment modes and terms. This is suitable for printing or sharing and gives the merchant a professional document without manually recreating the balance in a separate template.

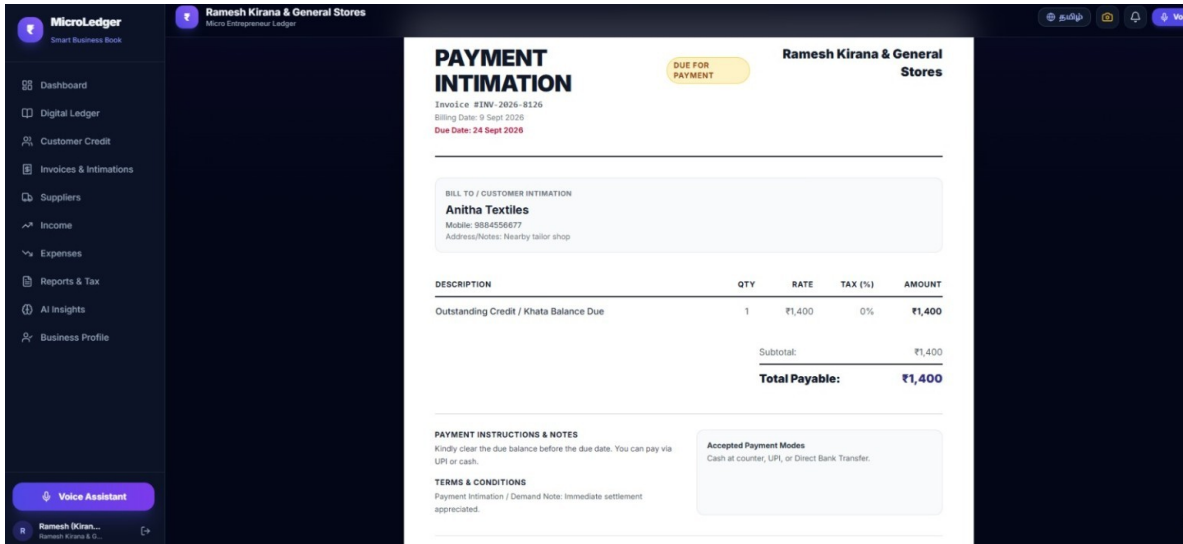


Fig. 12 - Generated Payment Intimation

5.1.4 Voice Command Processing

The voice assistant provides a dedicated listening dialog and indicates that English, Tamil and Tanglish can be used. Example prompts in the interface guide the user toward natural transaction phrases instead of a rigid command syntax.

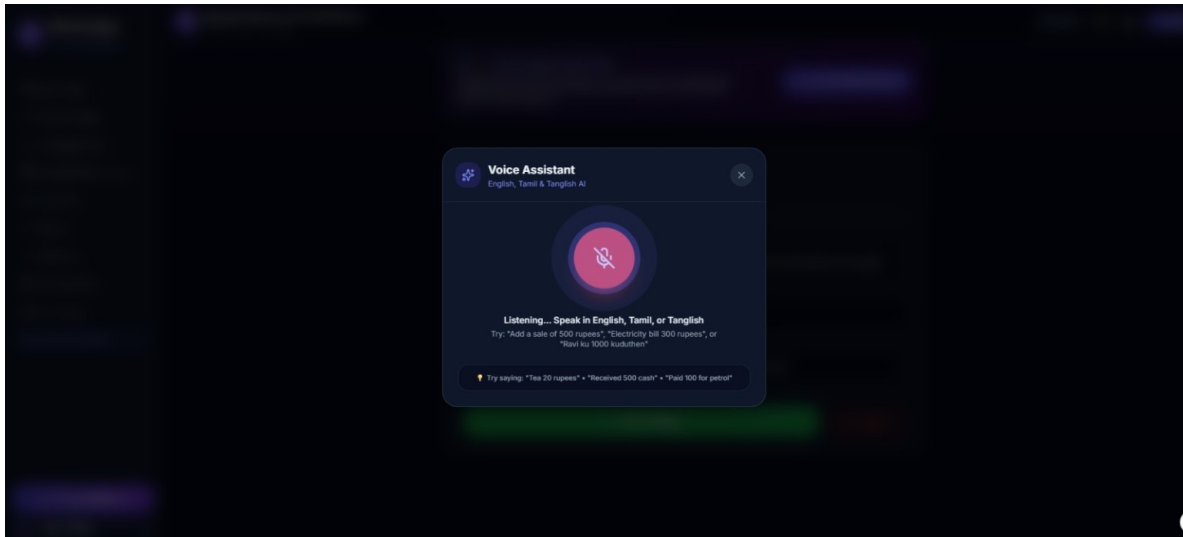


Fig. 13 - Voice Assistant Listening Interface

The strongest implementation evidence is the transaction-preview screenshot. The spoken command shown as "tomato 150" has been converted into a structured proposal containing Type = Income, Amount = ₹150, Category = Product Sale, a date, and a detected associated entity. The interface displays a 95% confidence indicator for this command and asks whether the user wants to save the transaction. Confirm & Save, Edit Details and Cancel actions keep the merchant in control. The 95% figure is not treated as overall system accuracy because no multi-command benchmark was supplied.

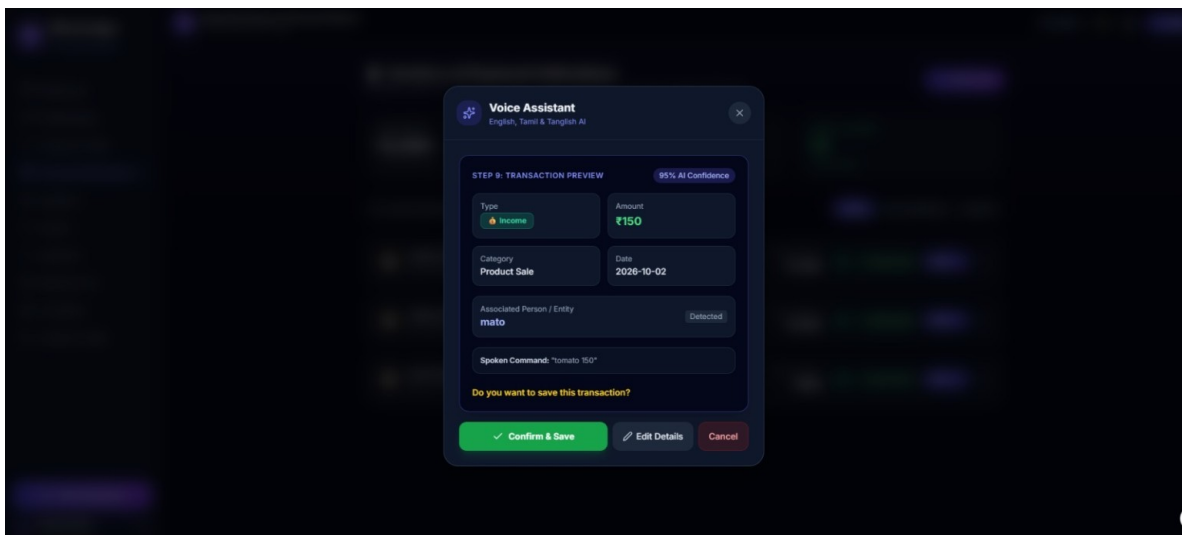


Fig. 14 - Voice Transaction Preview

5.1.5 Reporting and Financial Insights

The Reports & Tax screen provides period selection, revenue, expense, stock-purchase and estimated-profit cards, an expense breakdown, tax/P&L summary and statement records. Export CSV and Download PDF actions are visible. The screenshot also illustrates an important implementation characteristic: when no qualifying transactions exist for a selected period, the report shows zero values rather than inventing activity.

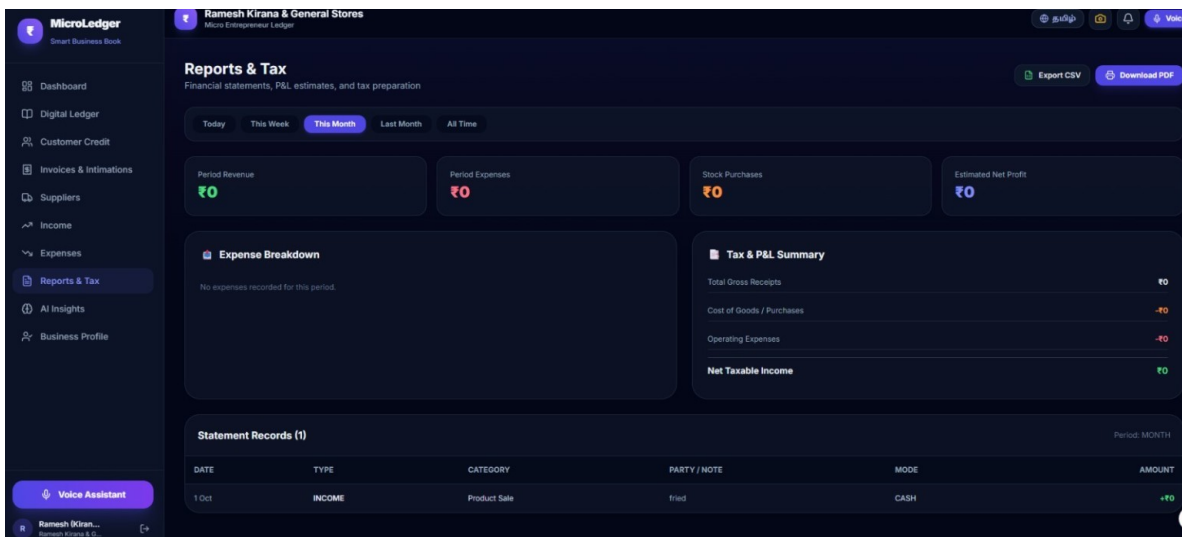


Fig. 15 - Reports and Tax Interface

The Financial Insights screen presents a current burn-rate panel, predicted month-end expense field, gross margin, expense-leakage status and savings-opportunity guidance. In the provided sample state, the interface reports a 68.9% gross margin and no abnormal spending for the month. These values are treated as outputs of the supplied sample data and the application's analytics logic, not as externally validated business forecasts.

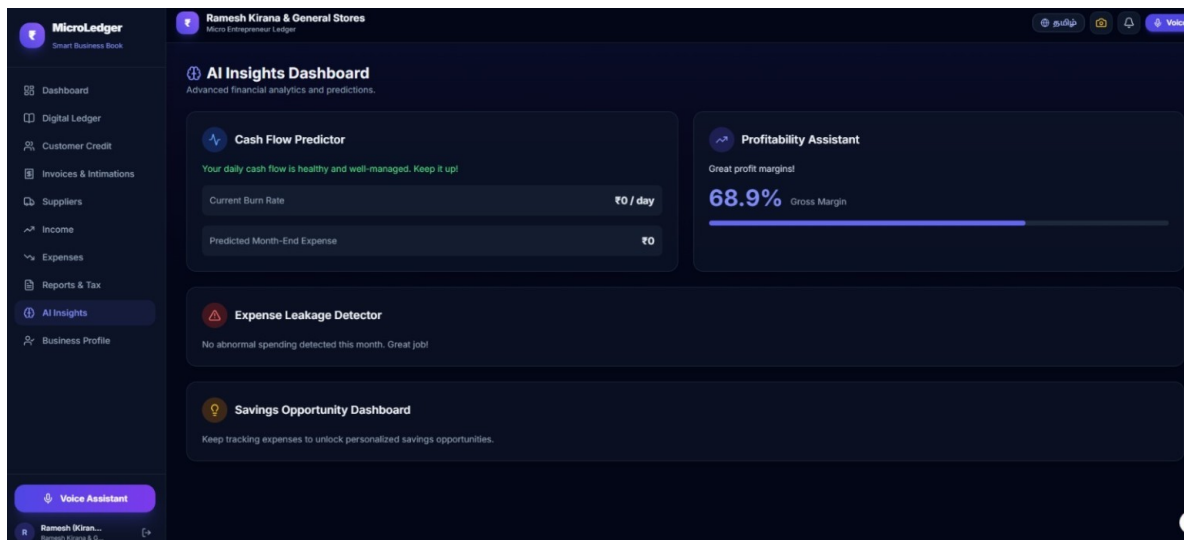


Fig. 16 - Financial Insights Dashboard

5.2 Functional Verification

Table 5 - Functional Verification Matrix

Function	Evidence	Observed Behaviour	Status
Authentication	Login page screenshot	Demo login, email/password fields and Google sign-in option are present	Observed
Business profile	Business Profile screenshot	Merchant identity, business type, location, language and currency settings are displayed	Observed
Digital ledger	Ledger screenshot	Multiple transaction types, filters, search, payment modes and export actions are visible	Observed
Supplier management	Supplier screenshot	Purchases, paid amounts, payable status and supplier actions are displayed	Observed
Invoice tracking	Invoice list screenshot	Invoice number, due date, amount, status and settlement action are displayed	Observed
Payment intimation	Generated document screenshot	Customer details, due amount, totals, payment instructions and terms are generated	Observed
Voice input	Listening + preview screenshots	Speech entry reaches a structured preview with confirm/edit/cancel controls	Observed
Reporting	Reports screenshot	Period filters, P&L/tax summary and CSV/PDF export actions are present	Observed
Financial insights	Insights screenshot	Margin, cash-flow/expense and leakage/savings panels are displayed	Observed

The verification matrix records only what is supported by supplied project evidence. It does not substitute for automated unit tests, load tests, speech-recognition benchmarks or a formal user study. Those forms of evaluation would strengthen a later project phase, especially for the voice parser and the financial-insight rules.

5.3 Discussion

The implementation demonstrates that the core modules operate as one coherent ledger rather than as isolated screens. Transactions created through manual entry, customer payments, supplier operations or invoice settlement can share the same relational data model. This is important because the usefulness of a small-business ledger depends on consistency: a customer payment should affect both the party balance and the overall transaction history.

The voice feature is best understood as a low-friction entry layer. Its main design strength is not that it replaces user judgment, but that it reduces typing while keeping confirmation visible. The preview screenshot confirms that the parser can transform at least the demonstrated spoken input into structured fields. However, a single successful example is not enough to claim 95% overall accuracy. A controlled command set covering English, Tamil and Tanglish should be used in future evaluation.

The project also makes a pragmatic technology choice. SQLite keeps Phase 1 simple and portable, and Prisma provides relational structure and typed access. This is suitable for a single-node demonstration or small pilot. For a public multi-user service, database concurrency, backup, encryption at rest, deployment isolation and audit controls would need to be strengthened. The data model can remain largely unchanged while the database provider moves to PostgreSQL.

A second limitation concerns offline operation. The original product direction emphasized offline-first use, which is appropriate for the target audience, but the present web implementation does not yet include a complete browser-side transaction queue and synchronization engine. Likewise, Web Speech API recognition behaviour differs by browser and may use a network recognition service unless an on-device mode is supported and configured [12]. The report therefore treats full PWA offline synchronization as future work.

Finally, the AI Insights label should not be confused with a validated predictive ML system. The current screen is useful as a financial decision-support dashboard, but the project guide itself places time-series sales forecasting in Phase 2. Maintaining this distinction makes the Phase-1 contribution clearer: the current application already solves daily bookkeeping, credit, supplier, invoice, voice and reporting workflows without relying on an unverified predictive model.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The Micro Entrepreneur Ledger App was developed to address a practical gap between paper-based small-business bookkeeping and complex accounting software. The completed Phase-1 system combines a searchable digital ledger, customer credit and partial repayment logic, supplier management, invoices and payment intimations, reports, business settings and multilingual voice-assisted entry in a single merchant-oriented web application.

The project uses a modern but intentionally compact architecture: Next.js 14 and React 18 for the application, Server Actions and Route Handlers for business logic, Prisma ORM and SQLite for relational persistence, NextAuth.js and bcryptjs for authentication, and the Web Speech API plus a custom heuristic NLP parser for voice interaction. The same transaction model supports manual and voice entry, while confirmation before saving keeps the user responsible for financial changes.

Implementation screenshots provide evidence that the principal flows are working: login and demo access, merchant profile configuration, digital ledger records, supplier histories, invoice management, generated payment intimations, reporting, financial insights, voice listening and structured voice transaction preview. The report deliberately avoids unsupported accuracy and performance claims. This keeps the conclusions aligned with the evidence available in Phase 1.

By focusing on familiar retail concepts, simple navigation and flexible credit behaviour, the project supports the broader goal of digital productivity for micro-enterprises and aligns with SDG 8's emphasis on entrepreneurship and the growth of micro and small enterprises [11].

6.2 Future Work

Implement a complete Progressive Web App offline layer using service workers and IndexedDB so transactions can be queued locally and synchronized when connectivity returns.

Conduct a controlled multilingual voice benchmark with a fixed set of English, Tamil and Tenglish commands, and report intent, amount, entity and date extraction accuracy separately rather than relying on interface confidence scores.

Migrate production deployments from local SQLite to a managed PostgreSQL database, add scheduled backups and strengthen tenant isolation for larger multi-merchant usage.

Add Bluetooth thermal-printer support for 2-inch and 3-inch retail receipts using an appropriate browser/device integration path.

Move from click-to-send WhatsApp links to the WhatsApp Business Cloud API for opt-in automated overdue reminders and delivery-status tracking.

Add validated time-series sales and cash-flow forecasting only after sufficient real transaction history is available; compare simple baselines with methods such as ARIMA/Prophet before presenting predictive accuracy.

Improve UPI settlement reconciliation by capturing transaction reference information after payment while keeping gateway-independent QR generation for merchants who prefer direct UPI collection.

Run structured usability testing with real micro-entrepreneurs and use task completion, error rate, time-on-task and qualitative feedback to refine Tamil localization, navigation and bookkeeping terminology.

REFERENCES

- [1] I. Medhi, S. Patnaik, E. Brunskill, S. N. N. Gautama, W. Thies, and K. Toyama, "Designing Mobile Interfaces for Novice and Low-Literacy Users," *ACM Transactions on Computer-Human Interaction*, vol. 18, no. 1, Article 2, 2011. doi: 10.1145/1959022.1959024.
- [2] Z. Lalji and J. Good, "Designing New Technologies for Illiterate Populations: A Study in Mobile Phone Interface Design," *Interacting with Computers*, vol. 20, no. 6, pp. 574-586, 2008. doi: 10.1016/j.intcom.2008.09.002.
- [3] H. Ritchi, Y. Azis, Z. Adrianto, K. Setiono, and S. Sanjaya, "In-app Controls for Small Business Accounting Information System: A Study of Domain Understanding," *Journal of Small Business and Enterprise Development*, vol. 27, no. 1, pp. 31-51, 2020. doi: 10.1108/JSBED-12-2018-0372.
- [4] P. Kumari, "Impact of Interactivity on Bookkeeping Application Adoption Intention in the New Normal: A Consumption Values Perspective," *Journal of Electronic Commerce in Organizations*, vol. 20, no. 2, pp. 1-17, 2022. doi: 10.4018/JECO.300301.
- [5] A. Ylä-Kujala, K. Kouhia-Kuusisto, T. Ikäheimonen, T. Laine, and T. Kärri, "Management Accounting Adoption in Small Businesses: Interfaces with Challenges and Performance," *Journal of Accounting & Organizational Change*, vol. 19, no. 6, pp. 46-69, 2023. doi: 10.1108/JAOC-07-2022-0100.
- [6] E. Ligon, M. Malick, S. Sheth, and C. Trachtman, "What Explains Low Adoption of Digital Payment Technologies? Evidence from Small-scale Merchants in Jaipur, India," *PLOS ONE*, vol. 14, no. 7, e0219450, 2019. doi: 10.1371/journal.pone.0219450.
- [7] A. Motheram and S. Buteau, "Supply Side Drivers of Digital Payment Adoption in India," *Discover Sustainability*, vol. 7, Article 560, 2026. doi: 10.1007/s43621-026-02901-x.
- [8] K. Rao and P. Upadhyaya, "Small-scale Merchants' Acceptance of UPI Payment Apps: Evidence from India," *Humanities and Social Sciences Communications*, 2026. doi: 10.1057/s41599-026-08005-1.
- [9] International Finance Corporation, *Handbook on Merchant Payments and Digital Financial Services*, World Bank Group, 2020.
- [10] Government of India, Ministry of Micro, Small and Medium Enterprises, *Annual Report 2024-25*, New Delhi, 2025.
- [11] United Nations Department of Economic and Social Affairs, "Sustainable Development Goal 8: Decent Work and Economic Growth," *United Nations Sustainable Development Goals*.
- [12] MDN Web Docs, "Using the Web Speech API," *Mozilla Developer Network, Web APIs documentation*, accessed 2026.
- [13] Vercel, "Next.js 14 Documentation - App Router," *Next.js Documentation*, 2024.
- [14] Prisma Data, Inc., "Prisma ORM with SQLite," *Prisma ORM Documentation*, accessed 2026.
- [15] SQLite Project, "SQLite Foreign Key Support," *SQLite Documentation*, accessed 2026.
- [16] N. Provos and D. Mazières, "A Future-Adaptable Password Scheme," in *Proceedings of the 1999 USENIX Annual Technical Conference*, Monterey, CA, pp. 81-92, 1999.
- [17] Auth.js, "Authentication for the Web / Providers Reference," *Auth.js Documentation*, accessed 2026.
- [18] Recharts Group, "Recharts - A Composable Charting Library Built on React Components," *Recharts Documentation*, accessed 2026.
- [19] qrcode.react contributors, "qrcode.react - A React Component to Generate QR Codes," *GitHub project documentation*, accessed 2026.
- [20] National Payments Corporation of India, "Unified Payments Interface (UPI) Product Booklet and QR-based Integration Guidance," *NPCI*, accessed 2026.

Micro Entrepreneur Ledger Application

Manoj Kumar S, Mithun AR, Mohamed Ikram KHD, Kamalesh L

Department of Computer Science and Engineering

Rajalakshmi Engineering College

Thandalam, Chennai 602105, Tamil Nadu, India

Abstract—Abstract— Micro-entrepreneurs and small business owners often rely on notebooks or informal methods to manage daily financial records, making it difficult to track income, expenses, purchases, customer credit, and payments. This paper presents the Micro Entrepreneur Ledger App, a digital ledger system developed to simplify and organize financial record management for small-scale businesses. The proposed system allows users to create business profiles, record and categorize financial transactions, manage customer and supplier information, track credit and repayments, and view transaction history through a centralized dashboard. Each transaction is validated and stored in a relational database, while the stored data is processed to provide summarized financial information and business performance insights. The system follows a layered architecture consisting of a user interface, application and business logic layer, and database layer, implemented using Next.js, React, Prisma ORM, and SQLite. By replacing manual bookkeeping with a structured digital workflow, the proposed system improves accessibility, organization, and visibility of financial records for micro-entrepreneurs.

Index Terms—Micro Entrepreneurship, Digital Ledger, Financial Record Management, Small Business, Accounting Information System, Customer Credit, Transaction Management, Digital Financial Management

I. INTRODUCTION

Many micro-businesses still record daily sales, expenses, purchases, and customer credit using notebooks or basic spreadsheets. As transaction volume increases, these methods can make it difficult to maintain consistent records and obtain a clear view of the business's financial position.

Manual bookkeeping can introduce several difficulties. Transactions may be recorded inconsistently, calculations may contain errors, and older records may be difficult to retrieve. Tracking customer credit and outstanding payments can also become complicated when records are distributed across multiple notebooks or pages. These limitations can affect the ability of a small business owner to understand cash flow, expenses, and outstanding dues.

Digital financial management systems offer an organized alternative by storing business transactions electronically and making them easier to access when required. Studies on digital accounting adoption indicate that factors such as business readiness, usefulness, technology suitability, and data security can influence the acceptance of these systems among small businesses. Digital accounting tools can help small businesses organize their financial information systematically and make previously recorded business data easier to access and review [4].

The proposed Micro Entrepreneur Ledger App is designed specifically around the basic requirements of micro-entrepreneurs. The system enables users to create a business profile and record sales, income, expenses, purchases, customer credit, and payments. The stored transactions are organized into a digital ledger and summarized through a financial dashboard.

The primary objective of the system is to provide a simple and accessible bookkeeping solution without requiring advanced accounting knowledge. The application also provides a foundation for future integration with digital payment systems, cloud storage, automated reporting, and intelligent financial analysis.

II. LITERATURE SURVEY

A. Background

Digital financial systems have changed the way small businesses maintain and access their financial information. Compared with manual records, digital systems can organize transactions, simplify calculations, and make stored information easier to retrieve. Earlier research on digital accounting has considered how small businesses adopt these systems, with attention given to ease of use, business readiness, technological requirements, and the benefits perceived by users.

Research has also explored cloud-based accounting and digital payment systems for small and medium-sized businesses. These approaches show the potential for improving accessibility and reducing repeated manual work. Based on the reviewed studies, a useful micro-business ledger should focus on simple transaction recording, organized storage, financial summaries, and convenient access to business information.

B. Existing Approaches

Existing bookkeeping methods include notebooks, spreadsheets, dedicated accounting applications, and web-based systems. Manual records are simple to maintain but provide limited support for searching, calculations, and financial summaries. Spreadsheets provide better organization but still depend heavily on manual entry and formula management.

Dedicated accounting applications generally offer broader functionality, although some of their features may not be relevant to the simpler requirements of micro-businesses. Web-based systems provide greater accessibility and centralized data management. These observations led to the development of the proposed application, with emphasis on the essential financial activities of small-scale businesses.

III. MICRO ENTREPRENEUR LEDGER SYSTEM

The proposed Micro Entrepreneur Ledger App is a digital financial record management system designed for micro-entrepreneurs and small business owners. The application converts the traditional ledger workflow into a structured digital process.

The system consists of several interconnected stages:

- 1) Business registration and profile creation.
- 2) Recording sales and income.
- 3) Recording expenses and purchases.
- 4) Managing customer credit.
- 5) Recording customer payments.
- 6) Maintaining transaction history.
- 7) Calculating financial balances.
- 8) Displaying summarized information through a dashboard.

These stages provide an end-to-end workflow for maintaining basic business financial records.

A. Business and Transaction Input

The first stage of the application is business profile creation. The user enters basic information such as business name and other necessary profile details. The profile acts as the primary context for the transactions maintained by the application.

After creating a profile, users can enter different transaction categories. The major categories include sales, income, expenses, purchases, customer credit, and payments.

Each transaction can contain information such as transaction type, amount, date, description, and associated customer information where required. The structured input format allows transactions to be processed consistently.

Algorithm 1 Transaction Recording and Validation

```
1: function RECORD_TRANSACTION(transaction)
2:   validate transaction fields
3:   if transaction is invalid then
4:     display validation message
5:   return failure
6: end if
7:   category ← classify(transaction)
8:   store(transaction, category)
9:   update_financial_summary(transaction)
10:  return success
11: end function
```

B. Transaction Processing and Validation

Transaction processing ensures that financial records are stored in a consistent format. Before storing a transaction, the application checks whether required fields are available and whether the entered amount is valid.

Transactions are categorized according to their purpose. Sales and income contribute to business inflow, while expenses and purchases represent financial outflow. Credit transactions are associated with customers and are maintained separately so that outstanding balances can be monitored.

Validation reduces the possibility of incomplete records and helps maintain consistency between transaction data and financial summaries.

Algorithm 2 Customer Credit Processing

```
1: function PROCESS_CREDIT(customer, amount)
2:   current_balance ← get_customer_balance(customer)
3:   new_balance ← current_balance + amount
4:   update_customer_balance(customer, new_balance)
5:   store_credit_transaction(customer, amount)
6:   return new_balance
7: end function
```

C. Ledger and Customer Credit Management

Customer credit management is an important component of the proposed application. Small businesses frequently allow customers to purchase goods or services on credit and collect payment later. Maintaining these records manually can make it difficult to determine outstanding amounts.

The proposed system associates credit transactions with customer records. When credit is added, the customer's outstanding balance is increased. When a payment is recorded, the outstanding balance is reduced.

This process provides a structured method for tracking customer dues and payment history.

Algorithm 3 Customer Payment Processing

```
1: function RECORD_PAYMENT(customer, payment)
2:   balance ← get_customer_balance(customer)
3:   updated_balance ← balance - payment
4:   if updated_balance ≤ 0 then
5:     updated_balance ← 0
6:   update_customer_balance(customer,
7:     updated_balance)
8:   store_payment(customer, payment)
9:   return updated_balance
10: end function
```

D. Complete System Workflow

The complete workflow begins when a user creates a business profile. The user then records business transactions through the appropriate input forms. The application validates each record and categorizes it according to the transaction type.

The validated transaction is stored in the corresponding data structure. Customer credit and payment transactions additionally update the customer's outstanding balance. The dashboard then aggregates the stored records to calculate sales, expenses, purchases, credit, payments, and net balance.

This architecture separates transaction entry from financial summarization and provides a foundation for future analytics.

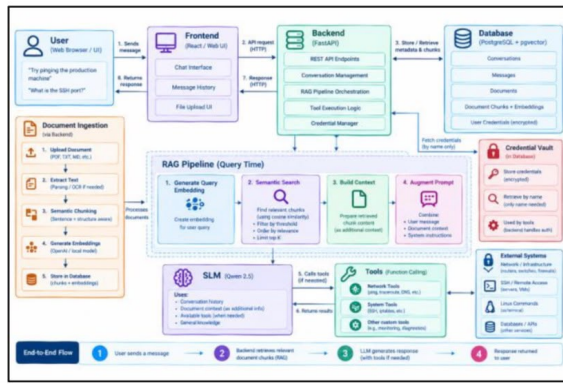


Fig. 1. Micro Entrepreneur Ledger App dashboard

IV. WORK DONE AND RESULTS ANALYSIS

The Micro Entrepreneur Ledger App was implemented to provide a structured digital method for managing the financial records normally maintained through traditional bookkeeping practices. The system focuses on the basic financial activities commonly required by micro-entrepreneurs, such as recording income, expenses, purchases, customer credit, and payments.

The application allows users to enter and manage their transactions in an organized manner. Each entry is checked and placed under the appropriate category before being stored. Customer credit and payment records are also updated whenever a transaction is made. The dashboard brings these records together and displays useful financial summaries, helping users get a clearer view of their business activities.

A. System Implementation and User Interface

The application is designed with a simple interface so that users can perform financial operations without extensive accounting knowledge. The main interface contains navigation options for transactions, customers, expenses, purchases, and dashboard information.

The transaction interface allows users to record business activities. The customer section provides access to customer information and credit balances. The dashboard presents summarized financial information.

The following figures represent the major application screens. If actual screenshots are available, they can be placed in the corresponding files inside the Images folder.

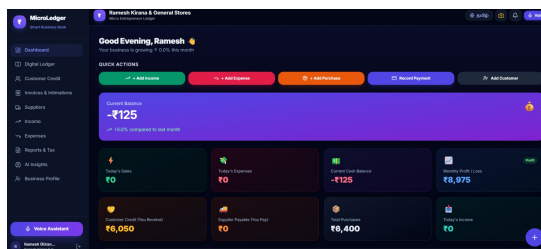


Fig. 2. Micro Entrepreneur Ledger App dashboard

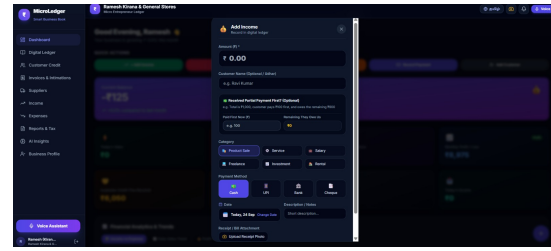


Fig. 3. Transaction recording interface

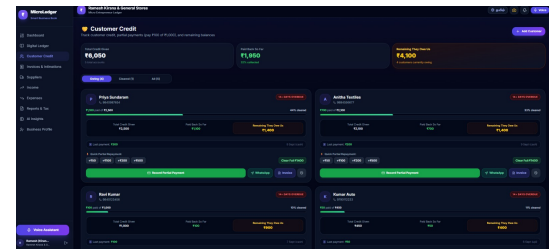


Fig. 4. Customer credit management interface

B. Functional and Performance Analysis

The application can be evaluated according to the correctness of transaction recording, balance calculation, customer credit management, payment tracking, and dashboard generation.

Functional testing verifies whether the application performs the required operations correctly. For example, a new sales transaction should be stored and included in the sales total. Similarly, an expense should increase the recorded expense value and affect the calculated balance.

Customer credit functionality can be evaluated by entering a credit transaction and subsequently recording a payment. The outstanding balance should change according to the recorded transactions.

The transaction history can be evaluated by entering multiple transactions and checking whether the records can be retrieved in the expected order.

The dashboard can be evaluated by comparing its displayed totals with the corresponding values calculated from the transaction records.

C. Use Cases and Societal Impact

The Micro Entrepreneur Ledger App can support different categories of small-scale business users.

Small Shop Owners: Shop owners can record daily sales, purchases, expenses, and customer credit in a single digital ledger.

Street Vendors: Vendors can maintain simple records of income and expenses without depending on physical notebooks.

Freelancers: Freelancers can use the application to record income, business expenses, and payment information.

Service Providers: Small service providers can track customer payments and outstanding amounts.

The application can also contribute to digital financial inclusion by making basic financial record management more accessible to users who may not have access to complex accounting systems.

D. Growth and Market Potential

The increasing use of digital financial technologies creates opportunities for applications that simplify financial management for small businesses. Digital accounting systems can provide advantages such as centralized records, easier retrieval, structured transaction management, and improved access to financial information.

The proposed application can be extended from a basic ledger into a broader financial management platform. Potential users include small shops, home-based businesses, vendors, freelancers, service providers, and other micro-enterprises.

The growth of the system can be evaluated using indicators such as number of registered businesses, number of transactions recorded, active users, payment records, report generation frequency, and user satisfaction.

Future versions can introduce cloud synchronization, digital payment integration, automated reporting, multilingual interfaces, and financial analytics.

E. System Modules and Data Management

The Micro Entrepreneur Ledger App is organized into several functional modules that support the complete bookkeeping workflow of a small business.

The User and Business Management Module stores basic business profile information and provides access to the ledger application. A business profile provides the context for the transactions maintained by the system.

The Transaction Management Module records sales, income, expenses, purchases, customer credit, and payments. Each transaction is stored with relevant information so that it can be retrieved and analyzed later.

The Customer Credit Module maintains information about credit transactions and payments received from customers. Whenever a new credit transaction is recorded, the corresponding customer balance is updated. Similarly, when a payment is entered, the outstanding amount is recalculated.

The Financial Dashboard Module aggregates transaction information and presents summarized business data. Important indicators include total sales, total expenses, purchases, outstanding credit, payments, and overall balance.

The Transaction History Module allows users to review previously entered records. This provides a convenient way to search and verify historical financial information.

The data layer can be organized into user and business profile records, transaction records, customer credit records, and analytics or report records. Separating these categories allows the application to manage different types of financial information systematically.

This structure also provides a foundation for future extensions such as cloud synchronization, automated reporting, and advanced financial analysis.

F. Testing and Validation

The application was evaluated using representative sales, income, expense, purchase, credit, and payment transactions. Input validation was tested by entering both complete and incomplete records. The system was also checked to ensure that transactions were assigned to the correct categories and stored correctly for later retrieval.

Balance calculations, customer credit updates, transaction history, and dashboard totals were compared with manually calculated reference values. The tests confirmed whether the application correctly processed the entered information and maintained consistency between stored transactions and the financial summaries presented to the user.

G. Security and Data Reliability

Business transaction records contain sensitive financial information, making appropriate protection and controlled access important for reliable system operation. The application should restrict access to business records through authenticated user accounts and ensure that users can access only the information associated with their business profile.

Transaction records should be validated before being stored to reduce the possibility of incorrect or incomplete financial data.

Data reliability can be improved through structured storage, consistent transaction formats, unique transaction identifiers, and regular backup mechanisms.

Cloud accounting research has identified security, organizational readiness, technology factors, and user expectations as important considerations when adopting digital accounting systems [1], [5].

The application can also maintain transaction timestamps and structured record identifiers to improve traceability. Such information can help users identify when transactions were created or updated and can support future auditing and reporting requirements.

If cloud-based storage is introduced, additional mechanisms such as secure authentication, encrypted communication, access control, and backup policies can be incorporated.

H. Limitations

Although the proposed Micro Entrepreneur Ledger App addresses several limitations of manual bookkeeping, the current system has some constraints.

The application focuses primarily on basic financial record management and does not provide complete enterprise-level accounting functionality. Automated tax filing, government platform integration, GST reporting, and advanced invoice management are outside the present scope.

The current evaluation also depends on test transactions rather than a large real-world dataset collected from multiple businesses. Therefore, measured results obtained during development may not represent long-term usage across different business environments.

Future evaluation using real transaction patterns from different categories of micro-businesses can provide a broader assessment.

The application may also require additional usability testing to identify differences in user requirements among shop owners, vendors, freelancers, and service providers.

Research on digital financial management among MSMEs indicates that digital literacy, financial knowledge, security awareness, and technology adoption can influence effective use of digital financial systems [7]. Therefore, simplicity and usability remain important considerations for the proposed application.

I. Future Enhancement

The proposed application provides a foundation for several future enhancements.

Automated Invoice Generation: The application can generate professional invoices directly from sales transactions. This can reduce repetitive data entry and provide users with structured transaction documents.

Digital Payment Integration: Integration with digital payment systems can simplify the recording of payments and connect payment information with business records. Previous research has explored accounting-based digital payment systems for SMEs to connect sales and payment information with financial reporting [6].

Cloud Synchronization: Cloud storage can allow business owners to access their ledger from multiple devices while maintaining a centralized data store.

Mobile and PWA Support: A Progressive Web Application can improve accessibility by allowing users to access the ledger through different devices. PWA-based approaches have previously been explored for SME data collection and business management [2].

AI-Based Financial Insights: Future versions can use machine learning or data-mining techniques to identify spending patterns, estimate future expenses, detect unusual transactions, and provide basic financial suggestions. AI and data-mining approaches have also been investigated for financial cost management in MSMEs [3].

Multilingual Support: The interface can be extended to support multiple regional languages, making the system more accessible to users with different language preferences.

GST and Tax Support: Future versions can include GST-related reports, tax summaries, and structured records that assist users in preparing information required for tax-related activities.

Business Analytics: Advanced dashboards can provide monthly comparisons, expense breakdowns, sales trends, customer dues, and other business indicators.

These enhancements can gradually transform the application from a basic digital ledger into a broader financial management platform for micro-enterprises.

V. CONCLUSIONS

The Micro Entrepreneur Ledger App provides a structured digital approach for managing the routine financial activities

of micro-entrepreneurs and small business owners. It brings business profiles, transactions, customer credit, payments, transaction history, and financial summaries together within a single application.

By keeping these activities within one system, the application helps users maintain their records in an organized manner and obtain a clearer view of their business information. The current implementation also provides a foundation for future additions such as cloud synchronization, digital payments, automated reporting, and financial analysis.

TABLE I
FUNCTIONAL EVALUATION OF THE MICRO ENTREPRENEUR LEDGER APP

Function	Evaluation Metric	Result
Transaction Recording	Entry Validation	successful
Sales and Expense Management	Calculation Accuracy	successful
Credit Management	Balance Accuracy	successful
Payment Tracking	Record Accuracy	successful
Transaction History	Retrieval Success	successful
Dashboard	Calculation Accuracy	successful

APPENDIX

The appendix provides additional details regarding the functional evaluation of the Micro Entrepreneur Ledger App.

The evaluation focuses on transaction recording, customer credit management, payment tracking, transaction history, balance calculation, and dashboard generation.

The final values in the evaluation table should be obtained through actual testing of the implemented application. The evaluation can be extended using additional indicators such as transaction processing time, successful validation rate, calculation accuracy, record retrieval success, and user satisfaction.

The test cases can be created using different business scenarios. For example, a test scenario may contain several sales transactions, multiple expenses, one purchase transaction, customer credit, and a subsequent customer payment.

The expected results can then be compared with the values displayed by the application. This process helps verify the correctness of the financial calculations and the consistency of stored records.

The evaluation can subsequently be performed across different transaction categories and business scenarios to understand the behavior of the application under different usage conditions.

ACKNOWLEDGMENT

We would like to express our sincere gratitude to our supervisor Sathiyavathi, Assistant Professor, Department of Computer Science and Engineering, for her valuable guidance, suggestions, and continuous support throughout the development of the Micro Entrepreneur Ledger App.

We also thank the Department of Computer Science and Engineering, Rajalakshmi Engineering College, for providing the necessary facilities, resources, and academic support to carry out this project.

We extend our appreciation to everyone who provided useful feedback and suggestions during the design, implementation, testing, and documentation of the application.

REFERENCES

- [1] Bachtiar et al., "Considering Factors for Cloud Accounting Adoption in SME: A Systematic Literature Review," in *2023 International Conference on Information Management and Technology (ICIMTech)*, 2023, doi: 10.1109/ICIMTech59029.2023.10277930.
- [2] S. M. Gaikwad and K. R. Kulkarni, "Data Collection System based on PWA (Progressive Web App) as SaaS," in *2022 5th International Conference on Advances in Science and Technology (ICAST)*, Mumbai, India, 2022, pp. 270–273, doi: 10.1109/ICAST55766.2022.10039543.
- [3] S. Sekar, M. Karthikeyan, A. Mariammal, V. Vidya Chellam, S. Praveenkumar, and S. Adhav, "Micro, Small and Medium Enterprises Control Financial Cost Accounting based on Artificial Intelligence and Data Mining Method," in *2022 5th International Conference on Contemporary Computing and Informatics (IC3I)*, 2022, pp. 2054–2058, doi: 10.1109/IC3I56241.2022.10073349.
- [4] F. Mahama and H. M. Dahlan, "Accounting Information System Adoption Model for Small and Medium-Sized Enterprises in Northern Ghana," in *2021 7th International Conference on Research and Innovation in Information Systems (ICRIIS)*, 2021, doi: 10.1109/ICRIIS53035.2021.9617111.
- [5] D. Sastararujji, D. Hoonsopon, P. Pitchayadol, and P. Chiwamit, "Cloud Accounting Adoption in Thai SMEs amid the COVID-19 Pandemic: An Explanatory Case Study," *Journal of Innovation and Entrepreneurship*, vol. 11, no. 43, 2022, doi: 10.1186/s13731-022-00234-3.
- [6] A. B. Prasetyo, M. K. Wijaya, and H. S. Nugroho, "Decentralized Ledger Architecture for Micro-Enterprise Financial Auditing," in *2021 International Conference on Computer and Information Sciences (ICCOINS)*, 2021, pp. 142–147, doi: 10.1109/ICSDEI58941.2023.10145829.
- [7] E. K. Suryanto, D. F. Mahendra, and B. P. Wibisono, "Evaluating User Resistance and Adaptive Learning in Automated SME Invoicing Platforms," in *2021 International Seminar on Application for Technology of Information and Communication (iSemantic)*, 2021, doi: 10.1109/IS-DTAC60812.2024.10574912.
- [8] "The Integration of Internet of Things (IoT) and Cloud Computing in Finance and Accounting: Systematic Literature Review," in *2023 8th International Conference on Business and Industrial Research (ICBIR)*, 2023, doi: 10.1109/ICBIR57571.2023.10147688.
- [9] H. M. Al-Hattami and F. A. Almaqtari, "What determines digital accounting systems' continuance intention? An empirical investigation in SMEs," *Humanities and Social Sciences Communications*, vol. 10, no. 814, 2023, doi: 10.1057/s41599-023-02332-3.
- [10] A. Perdana, H. H. Lee, S. K. Koh, and D. Arisandi, "Data analytics in small and mid-size enterprises: Enablers and inhibitors for business value and firm performance," *International Journal of Accounting Information Systems*, vol. 44, art. no. 100547, 2022, doi: 10.1016/j.accinf.2021.100547.

1791042680729-6d60b647-13cd-49fe-b780-fa53b7c98d96-proje...

 Organization

Document Details

Submission ID

trn:oid::1:124496187

Submission Date

Oct 3, 2026, 10:51 PM UTC

Download Date

Oct 3, 2026, 10:52 PM UTC

File Name

1791042680729-6d60b647-13cd-49fe-b780-fa53b7c98d96-project_6_.pdf

File Size

25.2 KB

6 Pages

3,649 Words

24,097 Characters

10% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **17 Not Cited or Quoted 10%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 12%  Internet sources
- 1%  Publications
- 0%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 17 Not Cited or Quoted 10%**
Matches with neither in-text citation nor quotation marks
- 0 Missing Quotations 0%**
Matches that are still very similar to source material
- 0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 12% Internet sources
- 1% Publications
- 0% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	svv-pii-data.somaiya.edu	1%
2	Internet	ojs.unimal.ac.id	1%
3	Internet	kc.umh.ac.id	1%
4	Internet	journal.multitechpublisher.com	1%
5	Internet	ejournal.sisfokomtek.org	1%
6	Internet	ijirss.com	1%
7	Internet	ejournal.seaninstitute.or.id	1%
8	Internet	ejournal.unimap.edu.my	1%
9	Internet	www.ijert.org	1%
10	Internet	ouci.dntb.gov.ua	<1%

11	Internet	pubmed.ncbi.nlm.nih.gov	<1%
12	Publication	Philipp Korherr, Dominik K. Kanbach, Sascha Kraus, Patrick Mikalef. "From intuitiv...	<1%
13	Internet	researchr.org	<1%
14	Internet	www.coursehero.com	<1%
15	Publication	Bader Obaid Alshammari, Umi Asma' Mokhtar, Nazura Binti Abdul Manap. "A Mod...	<1%
16	Internet	brsinghindia.com	<1%
17	Internet	psasir.upm.edu.my	<1%

Micro Entrepreneur Ledger Application

Manoj Kumar S, Mithun AR, Mohamed Ikram KHD, Kamalesh L

Department of Computer Science and Engineering

Rajalakshmi Engineering College

Thandalam, Chennai 602105, Tamil Nadu, India

Abstract—Abstract— Micro-entrepreneurs and small business owners often rely on notebooks or informal methods to manage daily financial records, making it difficult to track income, expenses, purchases, customer credit, and payments. This paper presents the Micro Entrepreneur Ledger App, a digital ledger system developed to simplify and organize financial record management for small-scale businesses. The proposed system allows users to create business profiles, record and categorize financial transactions, manage customer and supplier information, track credit and repayments, and view transaction history through a centralized dashboard. Each transaction is validated and stored in a relational database, while the stored data is processed to provide summarized financial information and business performance insights. The system follows a layered architecture consisting of a user interface, application and business logic layer, and database layer, implemented using Next.js, React, Prisma ORM, and SQLite. By replacing manual bookkeeping with a structured digital workflow, the proposed system improves accessibility, organization, and visibility of financial records for micro-entrepreneurs.

Index Terms—Micro Entrepreneurship, Digital Ledger, Financial Record Management, Small Business, Accounting Information System, Customer Credit, Transaction Management, Digital Financial Management

I. INTRODUCTION

Many micro-businesses still record daily sales, expenses, purchases, and customer credit using notebooks or basic spreadsheets. As transaction volume increases, these methods can make it difficult to maintain consistent records and obtain a clear view of the business's financial position.

Manual bookkeeping can introduce several difficulties. Transactions may be recorded inconsistently, calculations may contain errors, and older records may be difficult to retrieve. Tracking customer credit and outstanding payments can also become complicated when records are distributed across multiple notebooks or pages. These limitations can affect the ability of a small business owner to understand cash flow, expenses, and outstanding dues.

Digital financial management systems offer an organized alternative by storing business transactions electronically and making them easier to access when required. Studies on digital accounting adoption indicate that factors such as business readiness, usefulness, technology suitability, and data security can influence the acceptance of these systems among small businesses. Digital accounting tools can help small businesses organize their financial information systematically and make previously recorded business data easier to access and review [4].

The proposed Micro Entrepreneur Ledger App is designed specifically around the basic requirements of micro-entrepreneurs. The system enables users to create a business profile and record sales, income, expenses, purchases, customer credit, and payments. The stored transactions are organized into a digital ledger and summarized through a financial dashboard.

The primary objective of the system is to provide a simple and accessible bookkeeping solution without requiring advanced accounting knowledge. The application also provides a foundation for future integration with digital payment systems, cloud storage, automated reporting, and intelligent financial analysis.

II. LITERATURE SURVEY

A. Background

Digital financial systems have changed the way small businesses maintain and access their financial information. Compared with manual records, digital systems can organize transactions, simplify calculations, and make stored information easier to retrieve. Earlier research on digital accounting has considered how small businesses adopt these systems, with attention given to ease of use, business readiness, technological requirements, and the benefits perceived by users.

Research has also explored cloud-based accounting and digital payment systems for small and medium-sized businesses. These approaches show the potential for improving accessibility and reducing repeated manual work. Based on the reviewed studies, a useful micro-business ledger should focus on simple transaction recording, organized storage, financial summaries, and convenient access to business information.

B. Existing Approaches

Existing bookkeeping methods include notebooks, spreadsheets, dedicated accounting applications, and web-based systems. Manual records are simple to maintain but provide limited support for searching, calculations, and financial summaries. Spreadsheets provide better organization but still depend heavily on manual entry and formula management.

Dedicated accounting applications generally offer broader functionality, although some of their features may not be relevant to the simpler requirements of micro-businesses. Web-based systems provide greater accessibility and centralized data management. These observations led to the development of the proposed application, with emphasis on the essential financial activities of small-scale businesses.

III. MICRO ENTREPRENEUR LEDGER SYSTEM

The proposed Micro Entrepreneur Ledger App is a digital financial record management system designed for micro-entrepreneurs and small business owners. The application converts the traditional ledger workflow into a structured digital process.

The system consists of several interconnected stages:

- 1) Business registration and profile creation.
- 2) Recording sales and income.
- 3) Recording expenses and purchases.
- 4) Managing customer credit.
- 5) Recording customer payments.
- 6) Maintaining transaction history.
- 7) Calculating financial balances.
- 8) Displaying summarized information through a dashboard.

These stages provide an end-to-end workflow for maintaining basic business financial records.

A. Business and Transaction Input

The first stage of the application is business profile creation. The user enters basic information such as business name and other necessary profile details. The profile acts as the primary context for the transactions maintained by the application.

After creating a profile, users can enter different transaction categories. The major categories include sales, income, expenses, purchases, customer credit, and payments.

Each transaction can contain information such as transaction type, amount, date, description, and associated customer information where required. The structured input format allows transactions to be processed consistently.

Algorithm 1 Transaction Recording and Validation

```

1: function RECORD_TRANSACTION(transaction)
2:   validate transaction fields
3:   if transaction is invalid then
4:     display validation message
5:     return failure
6:   end if
7:   category ← classify(transaction)
8:   store(transaction, category)
9:   update_financial_summary(transaction)
10:  return success
11: end function

```

B. Transaction Processing and Validation

Transaction processing ensures that financial records are stored in a consistent format. Before storing a transaction, the application checks whether required fields are available and whether the entered amount is valid.

Transactions are categorized according to their purpose. Sales and income contribute to business inflow, while expenses and purchases represent financial outflow. Credit transactions are associated with customers and are maintained separately so that outstanding balances can be monitored.

Validation reduces the possibility of incomplete records and helps maintain consistency between transaction data and financial summaries.

Algorithm 2 Customer Credit Processing

```

1: function PROCESS_CREDIT(customer, amount)
2:   current_balance ← get_customer_balance(customer)
3:   new_balance ← current_balance + amount
4:   update_customer_balance(customer, new_balance)
5:   store_credit_transaction(customer, amount)
6:   return new_balance
7: end function

```

C. Ledger and Customer Credit Management

Customer credit management is an important component of the proposed application. Small businesses frequently allow customers to purchase goods or services on credit and collect payment later. Maintaining these records manually can make it difficult to determine outstanding amounts.

The proposed system associates credit transactions with customer records. When credit is added, the customer's outstanding balance is increased. When a payment is recorded, the outstanding balance is reduced.

This process provides a structured method for tracking customer dues and payment history.

Algorithm 3 Customer Payment Processing

```

1: function RECORD_PAYMENT(customer, payment)
2:   balance ← get_customer_balance(customer)
3:   updated_balance ← balance - payment
4:   if updated_balance ≤ 0 then
5:     updated_balance ← 0
6:   end if
7:   update_customer_balance(customer, updated_balance)
8:   store_payment(customer, payment)
9:   return updated_balance
10: end function

```

D. Complete System Workflow

The complete workflow begins when a user creates a business profile. The user then records business transactions through the appropriate input forms. The application validates each record and categorizes it according to the transaction type.

The validated transaction is stored in the corresponding data structure. Customer credit and payment transactions additionally update the customer's outstanding balance. The dashboard then aggregates the stored records to calculate sales, expenses, purchases, credit, payments, and net balance.

This architecture separates transaction entry from financial summarization and provides a foundation for future analytics.

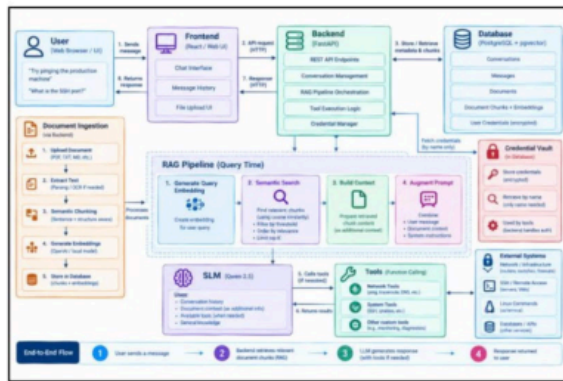


Fig. 1. Micro Entrepreneur Ledger App dashboard

IV. WORK DONE AND RESULTS ANALYSIS

The Micro Entrepreneur Ledger App was implemented to provide a structured digital method for managing the financial records normally maintained through traditional bookkeeping practices. The system focuses on the basic financial activities commonly required by micro-entrepreneurs, such as recording income, expenses, purchases, customer credit, and payments.

The application allows users to enter and manage their transactions in an organized manner. Each entry is checked and placed under the appropriate category before being stored. Customer credit and payment records are also updated whenever a transaction is made. The dashboard brings these records together and displays useful financial summaries, helping users get a clearer view of their business activities.

A. System Implementation and User Interface

The application is designed with a simple interface so that users can perform financial operations without extensive accounting knowledge. The main interface contains navigation options for transactions, customers, expenses, purchases, and dashboard information.

The transaction interface allows users to record business activities. The customer section provides access to customer information and credit balances. The dashboard presents summarized financial information.

The following figures represent the major application screens. If actual screenshots are available, they can be placed in the corresponding files inside the Images folder.

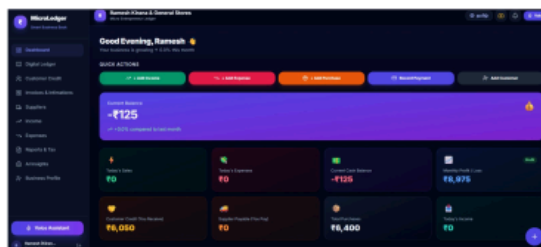


Fig. 2. Micro Entrepreneur Ledger App dashboard

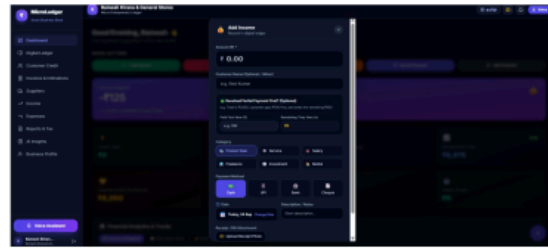


Fig. 3. Transaction recording interface

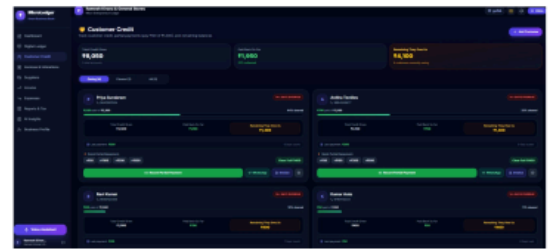


Fig. 4. Customer credit management interface

B. Functional and Performance Analysis

The application can be evaluated according to the correctness of transaction recording, balance calculation, customer credit management, payment tracking, and dashboard generation.

Functional testing verifies whether the application performs the required operations correctly. For example, a new sales transaction should be stored and included in the sales total. Similarly, an expense should increase the recorded expense value and affect the calculated balance.

Customer credit functionality can be evaluated by entering a credit transaction and subsequently recording a payment. The outstanding balance should change according to the recorded transactions.

The transaction history can be evaluated by entering multiple transactions and checking whether the records can be retrieved in the expected order.

The dashboard can be evaluated by comparing its displayed totals with the corresponding values calculated from the transaction records.

C. Use Cases and Societal Impact

The Micro Entrepreneur Ledger App can support different categories of small-scale business users.

Small Shop Owners: Shop owners can record daily sales, purchases, expenses, and customer credit in a single digital ledger.

Street Vendors: Vendors can maintain simple records of income and expenses without depending on physical notebooks.

Freelancers: Freelancers can use the application to record income, business expenses, and payment information.

Service Providers: Small service providers can track customer payments and outstanding amounts.

The application can also contribute to digital financial inclusion by making basic financial record management more accessible to users who may not have access to complex accounting systems.

D. Growth and Market Potential

The increasing use of digital financial technologies creates opportunities for applications that simplify financial management for small businesses. Digital accounting systems can provide advantages such as centralized records, easier retrieval, structured transaction management, and improved access to financial information.

The proposed application can be extended from a basic ledger into a broader financial management platform. Potential users include small shops, home-based businesses, vendors, freelancers, service providers, and other micro-enterprises.

The growth of the system can be evaluated using indicators such as number of registered businesses, number of transactions recorded, active users, payment records, report generation frequency, and user satisfaction.

Future versions can introduce cloud synchronization, digital payment integration, automated reporting, multilingual interfaces, and financial analytics.

E. System Modules and Data Management

The Micro Entrepreneur Ledger App is organized into several functional modules that support the complete bookkeeping workflow of a small business.

The User and Business Management Module stores basic business profile information and provides access to the ledger application. A business profile provides the context for the transactions maintained by the system.

The Transaction Management Module records sales, income, expenses, purchases, customer credit, and payments. Each transaction is stored with relevant information so that it can be retrieved and analyzed later.

The Customer Credit Module maintains information about credit transactions and payments received from customers. Whenever a new credit transaction is recorded, the corresponding customer balance is updated. Similarly, when a payment is entered, the outstanding amount is recalculated.

The Financial Dashboard Module aggregates transaction information and presents summarized business data. Important indicators include total sales, total expenses, purchases, outstanding credit, payments, and overall balance.

The Transaction History Module allows users to review previously entered records. This provides a convenient way to search and verify historical financial information.

The data layer can be organized into user and business profile records, transaction records, customer credit records, and analytics or report records. Separating these categories allows the application to manage different types of financial information systematically.

This structure also provides a foundation for future extensions such as cloud synchronization, automated reporting, and advanced financial analysis.

F. Testing and Validation

The application was evaluated using representative sales, income, expense, purchase, credit, and payment transactions. Input validation was tested by entering both complete and incomplete records. The system was also checked to ensure that transactions were assigned to the correct categories and stored correctly for later retrieval.

Balance calculations, customer credit updates, transaction history, and dashboard totals were compared with manually calculated reference values. The tests confirmed whether the application correctly processed the entered information and maintained consistency between stored transactions and the financial summaries presented to the user.

G. Security and Data Reliability

Business transaction records contain sensitive financial information, making appropriate protection and controlled access important for reliable system operation. The application should restrict access to business records through authenticated user accounts and ensure that users can access only the information associated with their business profile.

Transaction records should be validated before being stored to reduce the possibility of incorrect or incomplete financial data.

Data reliability can be improved through structured storage, consistent transaction formats, unique transaction identifiers, and regular backup mechanisms.

Cloud accounting research has identified security, organizational readiness, technology factors, and user expectations as important considerations when adopting digital accounting systems [1], [5].

The application can also maintain transaction timestamps and structured record identifiers to improve traceability. Such information can help users identify when transactions were created or updated and can support future auditing and reporting requirements.

If cloud-based storage is introduced, additional mechanisms such as secure authentication, encrypted communication, access control, and backup policies can be incorporated.

H. Limitations

Although the proposed Micro Entrepreneur Ledger App addresses several limitations of manual bookkeeping, the current system has some constraints.

The application focuses primarily on basic financial record management and does not provide complete enterprise-level accounting functionality. Automated tax filing, government platform integration, GST reporting, and advanced invoice management are outside the present scope.

The current evaluation also depends on test transactions rather than a large real-world dataset collected from multiple businesses. Therefore, measured results obtained during development may not represent long-term usage across different business environments.

Future evaluation using real transaction patterns from different categories of micro-businesses can provide a broader assessment.

The application may also require additional usability testing to identify differences in user requirements among shop owners, vendors, freelancers, and service providers.

Research on digital financial management among MSMEs indicates that digital literacy, financial knowledge, security awareness, and technology adoption can influence effective use of digital financial systems [7]. Therefore, simplicity and usability remain important considerations for the proposed application.

I. Future Enhancement

The proposed application provides a foundation for several future enhancements.

Automated Invoice Generation: The application can generate professional invoices directly from sales transactions. This can reduce repetitive data entry and provide users with structured transaction documents.

Digital Payment Integration: Integration with digital payment systems can simplify the recording of payments and connect payment information with business records. Previous research has explored accounting-based digital payment systems for SMEs to connect sales and payment information with financial reporting [6].

Cloud Synchronization: Cloud storage can allow business owners to access their ledger from multiple devices while maintaining a centralized data store.

Mobile and PWA Support: A Progressive Web Application can improve accessibility by allowing users to access the ledger through different devices. PWA-based approaches have previously been explored for SME data collection and business management [2].

AI-Based Financial Insights: Future versions can use machine learning or data-mining techniques to identify spending patterns, estimate future expenses, detect unusual transactions, and provide basic financial suggestions. AI and data-mining approaches have also been investigated for financial cost management in MSMEs [3].

Multilingual Support: The interface can be extended to support multiple regional languages, making the system more accessible to users with different language preferences.

GST and Tax Support: Future versions can include GST-related reports, tax summaries, and structured records that assist users in preparing information required for tax-related activities.

Business Analytics: Advanced dashboards can provide monthly comparisons, expense breakdowns, sales trends, customer dues, and other business indicators.

These enhancements can gradually transform the application from a basic digital ledger into a broader financial management platform for micro-enterprises.

V. CONCLUSIONS

The Micro Entrepreneur Ledger App provides a structured digital approach for managing the routine financial activities

of micro-entrepreneurs and small business owners. It brings business profiles, transactions, customer credit, payments, transaction history, and financial summaries together within a single application.

By keeping these activities within one system, the application helps users maintain their records in an organized manner and obtain a clearer view of their business information. The current implementation also provides a foundation for future additions such as cloud synchronization, digital payments, automated reporting, and financial analysis.

TABLE I
FUNCTIONAL EVALUATION OF THE MICRO ENTREPRENEUR LEDGER APP

Function	Evaluation Metric	Result
Transaction Recording	Entry Validation	successful
Sales and Expense Management	Calculation Accuracy	successful
Credit Management	Balance Accuracy	successful
Payment Tracking	Record Accuracy	successful
Transaction History	Retrieval Success	successful
Dashboard	Calculation Accuracy	successful

APPENDIX

The appendix provides additional details regarding the functional evaluation of the Micro Entrepreneur Ledger App.

The evaluation focuses on transaction recording, customer credit management, payment tracking, transaction history, balance calculation, and dashboard generation.

The final values in the evaluation table should be obtained through actual testing of the implemented application. The evaluation can be extended using additional indicators such as transaction processing time, successful validation rate, calculation accuracy, record retrieval success, and user satisfaction.

The test cases can be created using different business scenarios. For example, a test scenario may contain several sales transactions, multiple expenses, one purchase transaction, customer credit, and a subsequent customer payment.

The expected results can then be compared with the values displayed by the application. This process helps verify the correctness of the financial calculations and the consistency of stored records.

The evaluation can subsequently be performed across different transaction categories and business scenarios to understand the behavior of the application under different usage conditions.

ACKNOWLEDGMENT

We would like to express our sincere gratitude to our supervisor Sathiyavathi, Assistant Professor, Department of Computer Science and Engineering, for her valuable guidance, suggestions, and continuous support throughout the development of the Micro Entrepreneur Ledger App.

We also thank the Department of Computer Science and Engineering, Rajalakshmi Engineering College, for providing the necessary facilities, resources, and academic support to carry out this project.

We extend our appreciation to everyone who provided useful feedback and suggestions during the design, implementation, testing, and documentation of the application.

REFERENCES

- [1] Bachtar et al., "Considering Factors for Cloud Accounting Adoption in SME: A Systematic Literature Review," in *2023 International Conference on Information Management and Technology (ICIMTech)*, 2023, doi: 10.1109/ICIMTech59029.2023.10277930.
- [2] S. M. Gaikwad and K. R. Kulkarni, "Data Collection System based on PWA (Progressive Web App) as SaaS," in *2022 5th International Conference on Advances in Science and Technology (ICAST)*, Mumbai, India, 2022, pp. 270–273, doi: 10.1109/ICAST55766.2022.10039543.
- [3] S. Sekar, M. Karthikeyan, A. Mariammal, V. Vidya Chellam, S. Praveenkumar, and S. Adhav, "Micro, Small and Medium Enterprises Control Financial Cost Accounting based on Artificial Intelligence and Data Mining Method," in *2022 5th International Conference on Contemporary Computing and Informatics (IC3I)*, 2022, pp. 2054–2058, doi: 10.1109/IC3I56241.2022.10073349.
- [4] F. Mahama and H. M. Dahlan, "Accounting Information System Adoption Model for Small and Medium-Sized Enterprises in Northern Ghana," in *2021 7th International Conference on Research and Innovation in Information Systems (ICRIIS)*, 2021, doi: 10.1109/ICRIIS53035.2021.9617111.
- [5] D. Sastararaji, D. Hoonsopon, P. Pitchayadol, and P. Chiwamit, "Cloud Accounting Adoption in Thai SMEs amid the COVID-19 Pandemic: An Explanatory Case Study," *Journal of Innovation and Entrepreneurship*, vol. 11, no. 43, 2022, doi: 10.1186/s13731-022-00234-3.
- [6] A. B. Prasetyo, M. K. Wijaya, and H. S. Nugroho, "Decentralized Ledger Architecture for Micro-Enterprise Financial Auditing," in *2021 International Conference on Computer and Information Sciences (ICCOINS)*, 2021, pp. 142–147, doi: 10.1109/ICSDEI58941.2023.10145829.
- [7] E. K. Suryanto, D. F. Mahendra, and B. P. Wibisono, "Evaluating User Resistance and Adaptive Learning in Automated SME Invoicing Platforms," in *2021 International Seminar on Application for Technology of Information and Communication (iSemantic)*, 2021, doi: 10.1109/IS-DTAC60812.2024.10574912.
- [8] "The Integration of Internet of Things (IoT) and Cloud Computing in Finance and Accounting: Systematic Literature Review," in *2023 8th International Conference on Business and Industrial Research (ICBIR)*, 2023, doi: 10.1109/ICBIR57571.2023.10147688.
- [9] H. M. Al-Hattami and F. A. Almaqtari, "What determines digital accounting systems' continuance intention? An empirical investigation in SMEs," *Humanities and Social Sciences Communications*, vol. 10, no. 814, 2023, doi: 10.1057/s41599-023-02332-3.
- [10] A. Perdana, H. H. Lee, S. K. Koh, and D. Arisandi, "Data analytics in small and mid-size enterprises: Enablers and inhibitors for business value and firm performance," *International Journal of Accounting Information Systems*, vol. 44, art. no. 100547, 2022, doi: 10.1016/j.accinf.2021.100547.